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Cite as: Phys. Fluids **32**, 076101 (2020); https://doi.org/10.1063/5.0012468
Submitted: 01 May 2020 • Accepted: 16 June 2020 • Published Online: 01 July 2020

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ABSTRACT

The mixing enhancement of a coaxial jet with a Mach 1.4 primary jet and sonic secondary jet, at different convective Mach numbers, is presented in this study. Rectangular tabs of aspect ratios ($AR = h/w$, where h and w are the tab height and width, respectively) 2.0 and 0.75 were employed to manipulate the primary and secondary jets, respectively. The primary jet ($C0$) and coaxial jet without control or manipulation ($C1$) are studied in order to decouple the effects of rectangular tabs on jet mixing. Four different tab configurations were studied, viz., tabs placed in the primary jet ($C2$), tabs placed in the secondary jet ($C3$), and tabs placed in both primary and secondary jets with the relative orientation between them of 0° ($C4$) and 90° ($C5$), to document the effect of tabs on mixing. The supersonic core length L_c^* of the manipulated jet was used as a measure to quantify the mixing performance of the manipulated jet. The secondary flow reduces the growth rate of the primary shear layer and elongates the supersonic core length. This study reveals that the manipulated jet emulates the characteristic of a non-circular jet. Primary tabs are highly effective in reducing the supersonic core length of the coaxial jet than secondary tabs, and hence, the jet mixing increases. Two different flow categories of the manipulated jet have been identified. The physical reason behind the observed jet mixing and flow categories have been presented based on arguments related to changes in the flow field and shock structure.

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NOMENCLATURE

AR = aspect ratio of the tab ($= h/w$)
a_p = speed of sound in the primary jet
a_s = speed of sound in the secondary jet
a_∞ = speed of sound in the ambient stream
B = bypass ratio ($\dot{m}_s/\dot{m}_p$)
D_p = exit diameter of the primary nozzle
D^* = throat diameter of the primary nozzle
D_s = exit diameter of the secondary nozzle
D_{eq} = mass flow rate equivalent diameter, $D_p\sqrt{(\dot{m}_p + \dot{m}_s)/(\dot{m}_p)}$
h = height of the tab
L_c = supersonic core length
L_c^* = non-dimensionalized supersonic core length by D_p
M_p = Mach number at the exit of the primary nozzle
M_s = Mach number at the exit of the secondary nozzle
$\bar{M}_c$ = average convective Mach numbers, $(U_p - U_s)/(a_p + a_s)$, $(U_s)/(a_s + a_\infty)$, $(U_p)/(a_p + a_\infty)$
M_{cps} = convective Mach number of the eddies in the primary jet to the secondary jet
$M_{cs\infty}$ = convective Mach number of the eddies in the secondary jet to the ambient jet
$M_{cp\infty}$ = convective Mach number of the eddies in the primary jet to the ambient jet
$\dot{m}_s$ = mass flow rate at the exit of the primary nozzle
$\dot{m}_p$ = mass flow rate at the exit of the secondary nozzle
p_{0p} = pressure at the exit of the primary nozzle
p_{0s} = pressure at the exit of the secondary nozzle
p_{0t} = pressure measured by the Pitot tube
p_a = atmospheric pressure
p_e = nozzle exit pressure
R_{ps} = velocity ratio of the secondary jet to the primary jet, U_s/U_p
Re_p = Reynolds number based on the primary jet velocity (U_p) and nozzle exit diameter (D_p)
S_{ps} = density ratio of the secondary jet to the primary jet, ρ_s/ρ_p
$S_{s\infty}$ = density ratio of the ambient jet to the secondary jet, ρ_∞/ρ_s
$S_{p\infty}$ = density ratio of the ambient jet to the primary jet, ρ_∞/ρ_p
T_p = temperature in the primary jet
T_s = temperature in the secondary jet
T_∞ = temperature in the ambient jet

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U_p = velocity at the exit of the primary nozzle
U_s = velocity at the exit of the secondary nozzle
U_{cp} = convective velocity of the eddies in the primary jet
U_{cs} = convective velocity of the eddies in the secondary jet
w = width of the tab
x = coordinate along the jet axis
y = coordinate normal to the jet axis
z = coordinate mutually perpendicular to both x and y axes
γ_p = specific heat ratio of the primary jet
γ_s = specific heat ratio of the secondary jet
ρ_p = density of the primary jet
ρ_s = density of the secondary jet
μ_p = dynamic viscosity of the primary jet
δ' = growth rate of jet shear layers
δ'/δ'_0 = normalized growth rates of coaxial jet shear layers
$\Delta L^*_{c,C1}$ = percentage reduction in the supersonic core length with respect to the natural jet $C1$
$\Delta L^*_{c,C2}$ = percentage reduction in the supersonic core length with respect to the manipulated jet $C2$

I. INTRODUCTION

Jet noise is one of the major noise contributors of the overall engine noise,[1] and hence, the reduction in jet noise has gained significant attention among researchers in the past few decades. The near-field noise and far-field noise of supersonic jets are much noisier than their subsonic counterparts due to the intense Mach wave radiation.[2] In addition to the Mach wave noise, supersonic jets operating at the off-design condition result in a quasi-periodic shock-cell pattern,[1,3] leading to another noise component termed shock associated noise.[4] The main source of the shock associated noise arises from the intense interaction between the shock front and the convected vorticity within the jet plume.[4]

The Mach wave radiation of supersonic jets can be reduced by the presence of co-flows. The principle of Mach wave elimination is to surround the jet exhaust with a layer of the co-flowing stream called the secondary flow/jet whose properties are significantly different from those of the main flow. As such, the primary eddies from the primary jet become subsonic with respect to the co-flow/secondary flow and the co-flow eddies are subsonic with respect to the ambient air stream.[5] In doing so, the co-axial jets are slightly quieter than their counterpart single equivalent jet.[6] The flow structure of a co-flow jet configuration is considerably complex because the surrounding jet drastically influences the inner jet characteristics. The mixing between the streams is connected with and controlled by the dynamics and interactions of the vortical structures that are present in the shear layers developing between the two jet streams, and between the surrounding jet and the ambient fluid. Interestingly, the noise reduction of about 9 dB was achieved when a Mach 1.5 primary jet was enveloped by a co-flow of sonic Mach number.[7] Papamoschou[2] investigated the Mach wave elimination of a Mach 1.5 jet using a co-flow arising from a diameter ratio $D_s/D_p = 1.7$, where D_s is the co-flow or secondary flow nozzle diameter. Papamoschou reported that a thin co-flow ($D_s/D_p = 1.7$) got consumers faster by entrainment and hence was less effective in eliminating Mach waves than a thick one. The mass and momentum transfer between the two streams of a co-flowing jet with very large secondary flows at various velocity ratios of the secondary jet to the primary jet (U_s/U_p) reveals that U_s/U_p is the principal variable parameter that determines the shape of the mixing region.[8] A similar observation was also reported by Willliams *et al.*,[9] who investigated the flow structure and acoustics of cold subsonic coaxial jets and argued that noise attenuates when the jet is surrounded by an annular flow of variable velocity.

Murakami and Papamoschou[10,11] investigated the effect of the subsonic secondary stream on a Mach 1.5 primary jet and presented experimental results on the mean flow development and potential core lengths of single and dual stream compressible air jets. In the coaxial arrangements, the secondary flow reduced the growth rate of the primary shear layer and elongated the primary potential core. As a result, the mass entrainment rate of the coaxial jet was less than that of a single jet. Later, Dahl and Papamoschou[12] and Papamoschou and Rostamimonjezi[13] argued that the flow exiting a convergent–divergent nozzle operating at off-design conditions exhibits a strong instability that causes mixing enhancement in the flow itself and destabilizes an adjacent flow. The conditions of the primary flow were kept fixed, and the velocity of the convergent–divergent secondary flow was increased from low subsonic to sonic to supersonic. As the secondary flow reached sonic conditions, the spreading rate was significantly enhanced. With the increasing supersonic velocity of the secondary flow, the spreading rate decreased from the enhanced state and returned to normal values. It was also observed that the convergent secondary nozzle never produced mixing enhancement at any pressure ratio; rather, it stabilized the primary flow, elongating the primary potential core. In fact, Bogey *et al.*[14] and Papamoschou and Rostamimonjezi[13] correlated that there is a strong similarity between the location of peak noise and the end of the primary core.

The core of a supersonic co-flow jet can be reduced by either active or passive control techniques. Several passive and active flow control techniques have been developed in the recent years to reduce the shock-associated noise of a jet.[15–18] The passive control is accomplished by simple modifications of the nozzle shape (e.g., grooves, chevrons, and tabs) and hence does not require any external power source. The basic concept behind passive control techniques is to generate streamwise vorticity in the jet stream, thereby enhancing the jet mixing. In contrast, the active control is accomplished by adding energy to the flow in order to excite flow instabilities, leading to jet mixing.[15] Murakami and Papamoschou[10] used an eccentric jet with exactly the same co-flow mass flow rate as the coaxial jet to manipulate the potential core of the supersonic jet. The maximum elongation of the potential core of a supersonic jet in the eccentric configuration was only 20% in comparison with 68% in the co-flow configuration. The eccentric configuration shows substantial improvement in mixing over the coaxial case and achieves an entrainment rate comparable to that of the single jet. Even though the eccentric configuration serves as a better configuration than the co-axial one, no reduction in the potential core was observed with respect to the single jet. Hence, searching for a control technique for reduction of the potential core length leads to a well known technique called the "tab."

The tab as a passive control technique has gained reasonable attention over the other control techniques due to its simple design, low cost, and assembly at the nozzle exit. Kweon *et al.*[19] reported

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that the tab or wire suppresses the screech tones and the broadband shock-associated noise as well as the overall sound pressure level of a Mach 2.0 jet. Furthermore, the wire device breaks the shock-cell structure downstream of the device and causes strong instability waves that propagate both upstream and downstream. It is also well known that the mixing of a jet is related to the noise characteristic of a jet. That is, the noise can be reduced if the mixing is increased. Many studies have focused on the mixing effectiveness of small tabs, placed at the nozzle exit on the axisymmetric jets.[20,21] Most notably, Bradbury and Khadem[20] found that two tabs placed diametrically opposite at the nozzle exit were most effective mixing promoters, resulting in a drastic reduction of the jet core length. Presumably, the zone responsible for the source of noise is reduced.[11,14] Investigations performed to document the optimum shape of the tab reveal that the triangular-shaped tab, referred to as a delta tab, worked just as well compared to the rectangular tab.[22–24] Zaman *et al.*[22] conducted a parametric study on the tab geometry to optimize the tab blockage (ratio of the projected area of the tab to the nozzle exit area) for jet mixing and presented data of engineering interest, such as thrust loss, increase in mixing, and reduction in noise. The thrust loss varied approximately linearly with the geometric blockage, say, for the geometric blockage of 2.6%, the thrust loss encountered is ~5%.

The present work is motivated by the ability of the secondary flow in a coaxial jet to suppress Mach wave radiation emanating from the primary flow. However, the secondary flow reduces the growth rate of the primary shear layer and elongates the primary potential core,[13] the zone identified as a source for noise.[13,14] Only quite a few studies have been reported in the open literature for coaxial jets with supersonic primary flows.[13,25–28] There are very few experimental data on the potential core lengths of either the primary or the secondary flows. Hence, there is a need for benchmarking of data on coaxial supersonic jets primarily owing to phenomena that govern mixing and noise generation. Furthermore, from the brief review on jets manipulated by the tab, it is also clear that the effect of rectangular tabs on jet control has been studied extensively for jet mixing applications. In fact, the potential core for the jet manipulated by tabs has been drastically reduced.[22,23] Hence, in order to accelerate the growth rate of the primary shear layer and reduce the primary potential core of the coaxial jet, we employ tabs in the primary and secondary jets. Favorably, tabs have also been shown to reduce the shock-associated noise in the off-design condition of a supersonic jet.[22] However, there has been limited study reported in the open literature on the jet mixing capability of coaxial jets using tabs, and mostly on a subsonic jet and not on a supersonic primary jet. Naturally, one may wonder whether the tab is a potential passive control technique for reducing the jet core and, hence, enhancing the jet mixing of a supersonic coaxial jet. If so, whether the tabs should be placed in the primary jet and/or secondary jets. In other words, what is the optimum position and orientation of the tabs in a coaxial jet. In order to address these issues, the present work embarks on a systematic experimental study of a coaxial jet with a Mach 1.4 primary jet and sonic secondary jet manipulated by rectangular tabs. To optimize the tab position and orientation, four different tab configurations were studied, viz., tabs placed in the primary jet ($C2$), tabs placed in the secondary jet ($C3$), and tabs placed in both primary and secondary jets with the relative orientation between them of 0° ($C4$) and 90° ($C5$), along with the natural coaxial jet ($C1$).

The present work pushes the envelope of understanding the effect of tab position and orientation on jet mixing in greater detail. Extensive Pitot measurements are performed along the jet centerline, and the jet stream has been visualized using the shadowgraph technique in the orthogonal planes of the manipulated jet. This paper is organized as follows. The experimental setup and measurement details are given in Sec. II, followed by the discussion of mixing characteristics of $C1$ in Sec. III. The effect of tab orientation on the mixing characteristics of the manipulated jet is presented in Sec. IV. Finally, this work is concluded in Sec. V.

II. EXPERIMENTAL DETAILS

The setup of the experimental program and the associated diagnostic techniques used are first described in this section. This is then followed by a discussion of the jet properties identified for measurements and the data reduction techniques used to accomplish these tasks. With this in mind, three sub-sections, two devoted to the experimental setup and model configuration and another to measurements and data reduction, are presented below.

A. Jet facility

Experiments for the present investigation were performed in the jet facility, the same as that used by Kumar *et al.*[23,29] Refer to Ref. 23 for details regarding the jet facility. Here, we brief the important features of the jet facility. A two-stage reciprocating compressor driven by a 112 kW three-phase induction motor delivers air at 0.17 m^3/s at a pressure of 3.5 MPa. The compressed air is then cooled by a cooling tower and then passed through a pre-filter consisting of porous stone candles to remove solid contaminants such as rust particles and oil droplets. Finally, an activated carbon filter is used for finer filtering of the air. The compressed air at 10 bars stored in storage tanks of volume 85 m^3 is fed to the settling chamber via pressure regulating and pressure relief valves. To reduce the flow disturbance caused by the control valve, a mixing length of 1 m is placed between the valve and the settling chamber and is connected to the settling chamber by a wide angle diffuser. The settling chamber houses two wire mesh screens to reduce the turbulence entering the chamber of size 300 mm inner diameter and 600 mm length. Pressure tapings are provided in the settling chamber to monitor the stagnation pressure. The test models are fixed at the end of the settling chamber by a slot holder arrangement, which is a short pipe-like protrusion with an embedded O-ring to prevent leakage (see Fig. 1 in Ref. 23).

B. Experimental model

The experimental model is a coaxial nozzle with a conical convergent–divergent circular inner nozzle and a conical convergent circular outer nozzle made of brass. The inner nozzle and the outer nozzle are referred to as the primary and the secondary nozzle, respectively. The primary nozzle has an inner diameter (D_p) and throat diameter (D^*) of 11.24 mm and 10.56 mm, respectively, with the lip thickness t_p of 3.2 mm. The area ratio (=nozzle throat area/nozzle exit area) corresponds to a design exit Mach number of 1.43 in the choked condition. Papamoschou[2] investigated the Mach wave elimination of a Mach 1.5 jet using a co-flow of Mach number 1.5 arising from a diameter ratio $D_s/D_p = 1.7$, where D_s is the

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co-flow or secondary flow nozzle diameter. Papamoschou reported that a thin co-flow (D_s/D_p = 1.7) got consumers faster by entrainment and hence was less effective in eliminating Mach waves than a thick one. Hence, in the present investigation, a thick co-flow was chosen such that the secondary nozzle has an inner diameter (D_s) of 25.7 mm, corresponding to D_s/D_p = 2.3. Note that the lip thickness of the secondary nozzle t_s is 5 mm. The annular gap between the primary and the secondary nozzles is 4.0 mm [Fig. 1(a)]. Both the nozzles are mounted on a common base plate of 12 mm thickness via threads [Fig. 1(b)]. The base plate has 25 small circular holes of diameter 2.5 mm placed at a periphery radius of 20 mm from the center of the base plate [Fig. 1(b)]. These small holes feed the secondary nozzle. The coaxial nozzle was fixed at the end of the settling chamber as discussed in Ref. 23. Since there is a possibility of the actual Mach number to differ from the design Mach number, the primary nozzle after calibration was found to deliver a Mach 1.4 jet with an uncertainty <0.2%. Refer to Ref. 23 for details regarding the nozzle calibration.

As both the nozzles are connected to the same settling chamber, the stagnation pressure at the entry of both the nozzles is the same. Indeed, the secondary nozzle is fed by a number of small holes, from each of which emerges a smaller jet at the entrance of the secondary nozzle. It was observed that at the exit of the secondary nozzle, only 60%–80% of the stagnation pressure at the inlet is recovered for p_{0p}/p_a = 2.5–6 (Table I and Fig. 2). This loss of stagnation pressure can be accounted for by the mixing of the smaller jets through the small circular holes at its entry and the additional friction due to the outer walls of the primary nozzle. As such, the velocity ratio of the secondary jet to the primary jet (U_s/U_p) corresponding to p_{0s}/p_{0p} is also shown in Fig. 2. Note that U_s and U_p are estimated from p_{0s}/p_a and p_{0p}/p_a, respectively, as given in Refs. 30 and 31. Eventually, the bypass ratio $B = \dot{m}_s/\dot{m}_p$ is around 1.9–2.5 for U_s/U_p = 0.72–0.96. The value of B is close to that reported for a Mach 1.5 jet by Papamoschou[2] and Papamoschou and Debiasi.[5] Unlike the work of Papamoschou and Rostamimonjezi,[13] U_s/U_p in the present investigation is fixed for a given M_p.

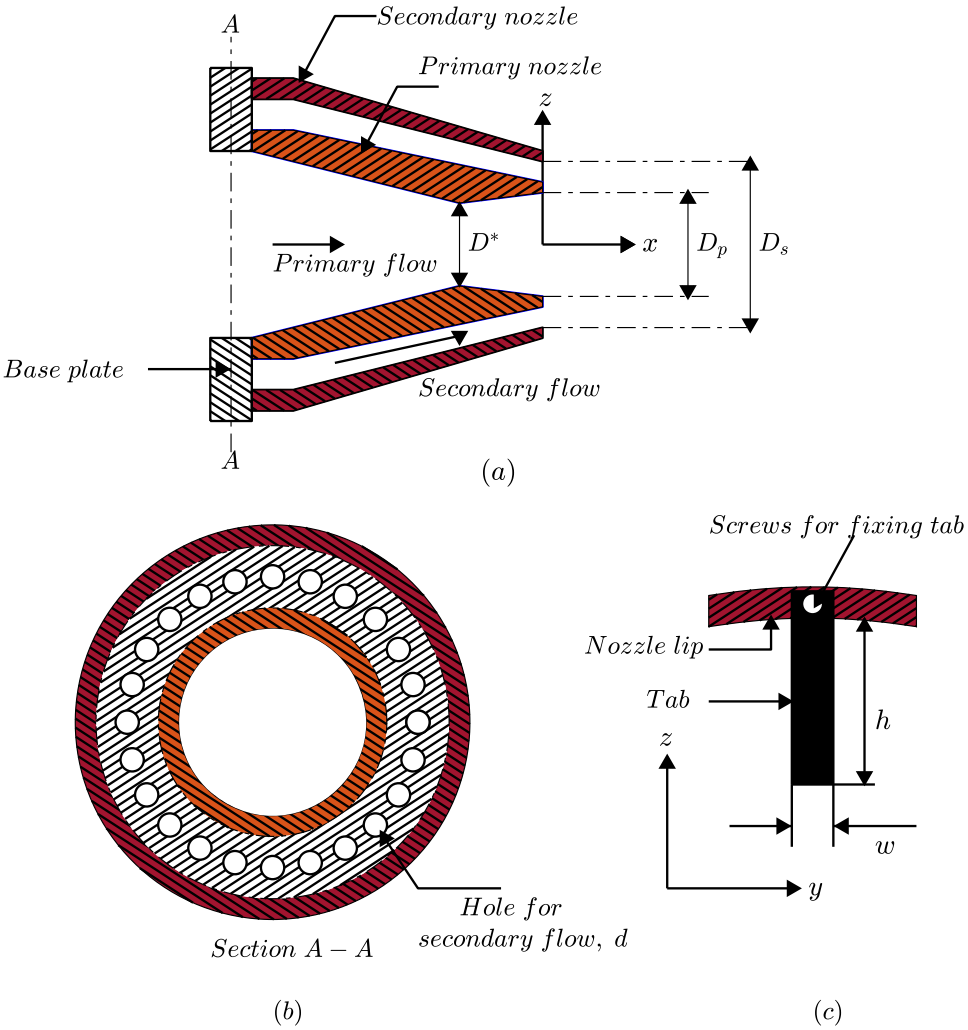


FIG. 1. (a) Schematic drawing of the coaxial jet nozzle: convergent–divergent primary nozzle and convergent secondary nozzle. (b) Detailed section $A-A$ of the base plate. (c) Tab dimensions in the yz-plane of tab height h and width w. Figure not to scale.

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TABLE I. Flow conditions.

p_{0p}/p_a	p_e/p_a	M_p	U_p (m/s)	p_{0s}/p_a	p_{0s}/p_{0p}	M_s	U_s (m/s)	U_s/U_p	M_{cps}	M_{cs_∞}
2.5	0.79	1.25	380	1.5	0.60	0.84	274	0.72	0.17	0.57
2.8	0.88	1.33	399	1.8	0.64	1.00	315	0.80	0.14	0.71
3.18	1.0	1.42	418	2.2	0.69	1.15	356	0.85	0.10	0.86
4.0	1.26	1.57	448	3.0	0.75	1.38	410	0.91	0.07	1.06
5.0	1.57	1.72	475	4.0	0.80	1.57	448	0.94	0.05	1.23
6.0	1.89	1.83	494	5.0	0.83	1.72	475	0.96	0.03	1.36

The secondary flow passes through a number of small holes before exiting the secondary nozzle; hence, one may wonder whether the flow exiting the secondary nozzle is uniform at the nozzle exit. Therefore, one has to ensure that all the small jets combine together to form a fully flowing uniform jet at the exit of the secondary nozzle. Figure 3 shows the variation of the pressure ratio (p_{0s}/p_{0p}) measured in the middle of the secondary flow [marked by a dotted line in Fig. 4(b)] for $p_{0s}/p_{0p} = 0.6$ and 0.83. It is explicitly seen that the flow exiting the secondary nozzle is uniform without any unduly oscillation in the radial direction. In other words, $p_{0s}/p_{0p} = 0.6$ is constant along the circumference in the middle of the secondary flow. In fact, a similar observation was seen for $p_{0s}/p_{0p} = 0.60$–0.83. The jet facility was able to run for about 30–50 min for the maximum and minimum test conditions as mentioned in Table I, which is sufficient enough to complete one set of measurements along the jet centerline or along the radial direction in the jet field.

Following Bradbury and Khadem[20] and Kumar *et al.*,[29] two sets of rectangular plain tabs were designed for the experiments and placed diametrically opposite to each other at the nozzle exit. Two sets of tabs, one for the primary flow and the other for the secondary flow, were used. Following Kumar *et al.*,[29] the total blockage of the tabs (=the tab projected area to the nozzle exit area) was chosen to be ~5%, i.e., a blockage of ~2.5% per tab of a set. It is worth mentioning that the inner area of the primary nozzle was used as the reference for the primary flow tabs and the area of the annulus between the primary and secondary nozzles was used as the reference for the secondary flow tabs. Since the intent of these mixing tabs is to accelerate mixing well in excess of the boundary layer region, Gretta and Smith[32] argued that the height of the mixing tab should be greater than the boundary layer thickness. Note that the boundary layer thickness of the Mach 1.4 nozzle was $0.04D_p$. Zaman[22] documented that the optimum increase in the jet entrainment with minimum thrust loss occurred for a size that corresponded to an area blockage of about 1.5%–2.5%. Hence, the height of the mixing tab should be greater than $0.04D_p$, and the tab blockage should be less than 2.5%. Since the annular gap between the primary and the secondary nozzles is 4.0 mm, the tab of height 2.2 mm was chosen for the secondary flows, and the reason is as follows. First, the present study focuses on the effect of primary and secondary tabs on jet mixing, given the same tab height. In fact, the distortion introduced by the primary and secondary tabs in the primary and secondary flows will be uniform, if the tab height is the same. Second, as the secondary flow arises from a relatively larger circumference than the primary flow, the width of the secondary tab has to be larger than that of the primary tab. As such, the tab aspect ratio AR (= h/w) would be different. Hence, we choose $AR = 2$ and 0.75 for the primary and secondary tabs, respectively. For the tab projected area of 2.5%, the width of the mixing tabs for $AR = 2$ and 0.75 was 1.1 mm and 3.0 mm, respectively. Concurrently, for approximately the same

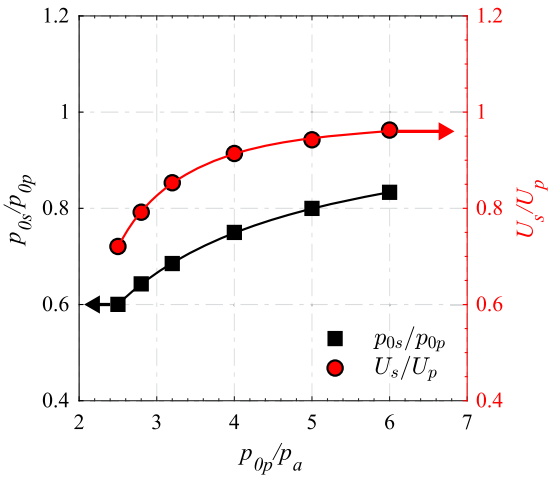


FIG. 2. Dependence of pressure recovery (p_{0s}/p_{0p}) and velocity ratio of the secondary jet to the primary jet (U_s/U_p) with p_{0p}/p_a.

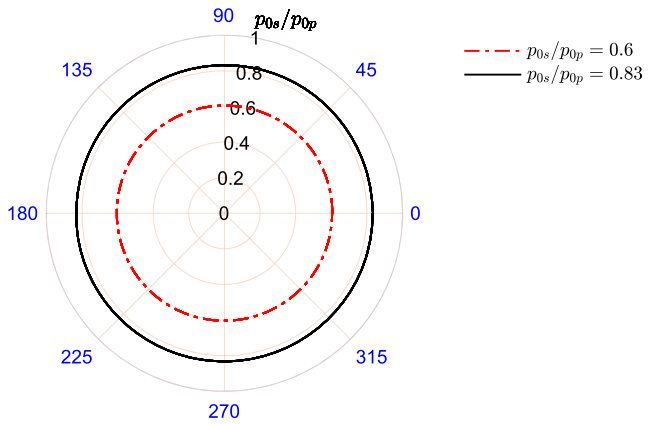


FIG. 3. Variation of pressure recovery (p_{0s}/p_{0p}) in the middle of the secondary flow [marked by a dotted line in Fig. 4(b)] for the minimum and maximum pressure conditions in Table I.

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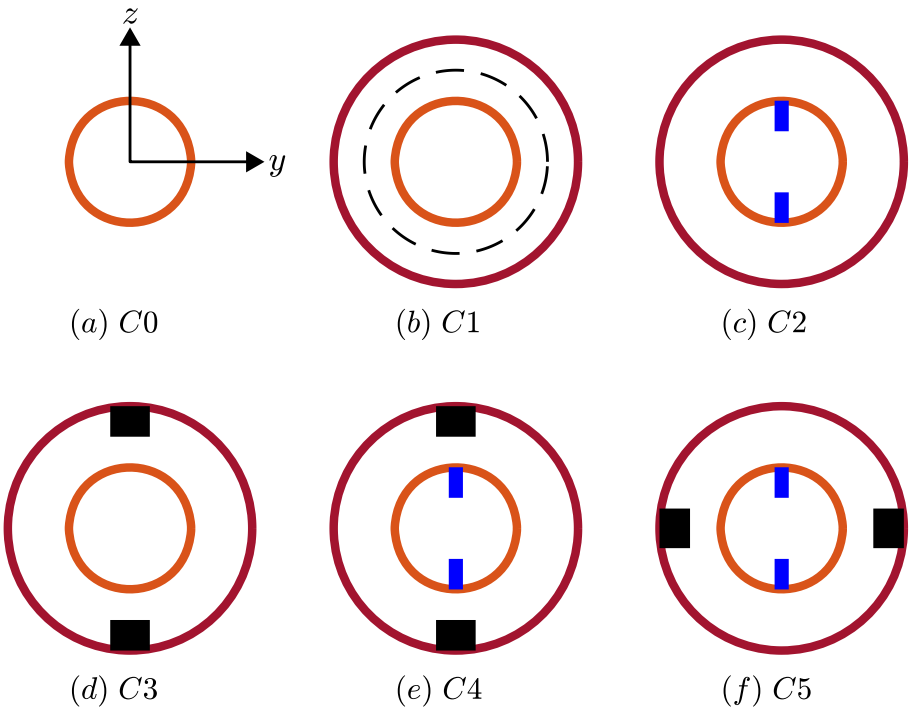


FIG. 4. Schematic diagram of the front view of the nozzle for the tabbed configurations. (a) Primary jet (*C*0). (b) Coaxial jet (*C*1). (c) Coaxial jet with the tab in a primary flow (*C*2). (d) Coaxial jet with the tab in a secondary flow (*C*3). (e) Coaxial jet with the tab in primary and secondary flows with 0° orientation (*C*4). (f) Coaxial jet with the tab in primary and secondary flows with 90° orientation (*C*5).

flow blockage, the distortion was greater when the tab width was larger as long as the height was larger than the boundary layer thickness.[33]

The primary flow tabs were fixed on the lip of the inner nozzle, directed toward the center of the nozzle. Since the inner nozzle lip thickness was only 2.65 mm, the primary flow tabs had to be pasted on the lip using an adhesive. Note that the tabs were intact with the nozzle during the experiments. The secondary flow tabs were fixed on the lip of the outer nozzle, directed toward the center of the nozzle, using screws to keep them in place over small holes drilled on the nozzle lip [Fig. 1(c)]. It is worth mentioning that introducing the tabs results in no change in p_{0s}/p_a or U_s/U_p at the nozzle exit. To characterize the effect of tab position and orientation, four different tab configurations were used in the present investigation, namely, tabs placed in the primary jet [*C*2, Fig. 4(c)], secondary jet [*C*3, Fig. 4(d)], and both primary and secondary jets with the relative orientation between them of 0° [*C*4, Fig. 4(e)] and 90° [*C*5, Fig. 4(f)]. The primary jet and coaxial jet without control [*C*0 and *C*1; Figs. 4(a) and 4(b)] were also studied in order to decouple the effects of the tabs on the jet.

C. Flow measurement and data processing

The pressure in the jet field was measured using a Pitot tube of 0.4 mm inner diameter and 0.6 mm outer diameter, fixed to a 3D traverse with a resolution of 0.1 mm in linear translation. Pressures were measured with a 16 channel Pressure Systems, Inc., 9010 transducer in the range of 0–2.1 MPa. The pressures were recorded at a sampling frequency of 250 Hz, for a time duration of 60 s, with a repeatable within ±2%. To visualize the shock-cell structure in a supersonic jet, the shadowgraph technique is used with a helium spark arc light source in conjunction with a concave mirror of 150 mm diameter. The light source was collimated by the condenser lens and was then brought to the concave mirror. The parallel beam from the mirror was made to pass through the jet flow field and projected on the screen. The surface finish of the mirror is $\lambda/6$, and the focal length is 1.6 m. Refer to Refs. 23 and 29 for further additional details on the instrumentation.

It is important to note that in a supersonic jet core, the Pitot probe measures the total pressure behind the bow-shock that stands ahead of the probe nose and not the actual total pressure. If the actual total pressure is required, then one has to correct the measured pressure for the pressure loss across the shock, which is practically impossible because of the complex shock-cell structure in the jet field. Even a conical supersonic probe will measure only the Mach number downstream of a shock that stands ahead of the probe nose and not the actual Mach number.[34] Nevertheless, the difference between the pressure measurement from a Pitot tube and the actual one is about less than 5%.[35] Hence, the data are accurate enough to capture the overall features, such as the supersonic core length, the number of shock cells, the spacing between them, and the jet spread.[29]

The coordinate system (x, y, z) is defined in Figs. 1 and 4. Measurements were conducted along the tab (xz) plane and the orthogonal or normal to the (xy) plane of the manipulated jet. Note that x^*, y^*, and z^* are the non-dimensionalized coordinates of x, y, and z, respectively, by D_p. The physical insight into the non-dimensionalization of x, y, and z by D_p is discussed in Sec. III. Experiments were conducted for the jet Reynolds numbers $Re_p = 3.5 * 10^5$ and $7 * 10^5$, respectively, with an uncertainty less than ±1%, for the minimum and maximum p_{0p}/p_a of 3 and 8 in the present investigation (Table I). The settling chamber pressure (p_{0s})

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was maintained constant during the experiments with an uncertainty less than ±1%. The pressure regulating valve was operated manually; therefore, fluctuations in the measured values of p_{0p} were less than ±1%. The experimental uncertainty of L_c^* was determined from repeated measurements and calibration of the Pitot tube. For the indirectly measured quantities such as $\Delta L^*_{c,C1}$, the method of propagation of uncertainties is used to determine their error bars, and the maximum uncertainties of $\Delta L^*_{c,C1}$ were found to be less than ±5%.

III. FLOW CHARACTERISTICS OF *C*1

In high-speed jets, the dominant source of noise is the Mach wave radiation.[36,37] Mach wave radiation is simply the Mach waves[2] generated by the supersonic motion of turbulent eddies in the jet field relative to the surrounding air. Surrounding the jet with a secondary flow at appropriate flow conditions may reduce the relative velocity of the eddies in the jet field, that is, the eddy velocity relative to the secondary flow becomes subsonic. This prevents the formation of shock waves. That is, the convective Mach number between the primary and secondary flows M_{cps} should be less than 1.0 to eliminate the Mach wave radiation. In fact, the upstream and cross-flow communication paths within the shear layer due to high Mach number are suppressed.[11] However, one must make sure that the secondary flow does not create Mach waves of its own with the ambient fluid, so its speed of the co-flow has an upper limit.[10] That is, the convective Mach number of the secondary flow $M_{cs\infty}$ should also be subsonic with respect to the ambient fluid. In other words, we require $M_{cps} < 1$ and $M_{cs\infty} < 1$.

Papamoschou and Bunyajitradulya[38] proposed an empirical model for the eddy convective velocity U_{cp}, based on its direct measurement in shear layers, which led to the appropriate choice of co-flow Mach number and temperature that satisfies $M_{cps} < 1$ and $M_{cs\infty} < 1$. Following Papamoschou,[2] one can readily derive relations between the co-flow static temperature ratio T_s/T_∞ and the co-flow Mach number M_s that satisfies $M_{cps} < 1$ and $M_{cs\infty} < 1$. Since those equations are given in Ref. 2, they are not provided here. Interested readers may refer to Eqs. (3)–(5) in Ref. 2. The dependence of M_s with T_s/T_∞ is plotted in Fig. 5 for $M_p = 1.25$ and 1.72. The lower and upper curves represent the condition at which $M_{cps} < 1$ and $M_{cs\infty} < 1$, respectively. It is evident that for subsonic M_s, the co-flow must be heated (large T_s/T_∞) for Mach waves to be eliminated. Charonko and Prestridge[39] studied the effect of the density ratio $\rho_0/\rho_\infty = 4.2$ and 1.2 on a co-flow jet to characterize the jet mixing. Their measurements on radial profiles and radial turbulent mass fluxes in the jet reveal that these quantities are seven times larger for $\rho_0/\rho_\infty = 4.2$ than for $\rho_0/\rho_\infty = 1.2$. As such, the shear layer growth for a hot jet is faster than that for a cold jet. That is, the low density at high effective temperature of the primary stream is similar to that observed in subsonic shear layers. However, the cold jets do not represent the actual scenario of an engine. The co-flow Mach numbers of the present investigation are marked by a symbol in Fig. 5. Note that M_{cps} and $M_{cs\infty}$ are obtained from Eq. (1) as given by Gutmark *et al.*[40] and Eqs. (2) and (3) as given by Papamoschou,[2] respectively,

$$M_{cps} = \frac{1}{1+\sqrt{S_{ps}}}\sqrt{\frac{\gamma_s}{\gamma_p}}M_s\left(\frac{1}{R_{ps}}-1\right), \tag{1}$$

$$M_{cs_\infty} = \bar{M}_c + \frac{d_{Mc}}{\sqrt{1+(a_s/a_\infty)^2}}, \tag{2}$$

$$d_{Mc} = \begin{cases} 1.5\bar{M}_c - 0.4, & \bar{M}_c > 0.27 \\ 0, & \bar{M}_c \le 0.27, \end{cases}$$

$$\bar{M}_c = \frac{U_s}{a_s + a_\infty}. \tag{3}$$

It is very clear for $M_p = 1.25$ [Fig. 5(a)] that the co-flow Mach number $M_s = 0.84$ results in Mach wave elimination. Indeed, $M_{cps} < 1$ and $M_{cs\infty} < 1$ as $M_{cps} = 0.17$ and $M_{cs\infty} = 0.41$. Hence, the present experimental data for $M_s = 0.84$ fall in between the two curves M_{cps} and $M_{cs\infty}$. On the other hand, the Mach wave is not eliminated for $M_p = 1.72$ [Fig. 5(b)] by the co-flow Mach number $M_s = 1.57$. Indeed, $M_{cps} < 1$ but $M_{cs\infty} > 1$ as explicitly seen from $M_{cps} = 0.05$ and $M_{cs\infty} = 1.14$. Hence, the present experimental data for $M_s = 1.57$ fall above $M_{cs\infty}$, highlighting the presence of Mach waves between the secondary flow and the ambient fluid. However, there is not Mach wave radiation between the primary and secondary jets. It is worth mentioning that the low M_{cps} is primarily due to large U_s/U_p (=0.94) as also observed by Gutmark *et al.*[40] A very low M_{cps} is related to U_s/U_p that is close to unity, thereby obtaining a high $M_{cs\infty}$ for relatively low U_∞/U_s. Note that in the present study, $U_\infty \approx 0$. Hence, there is an

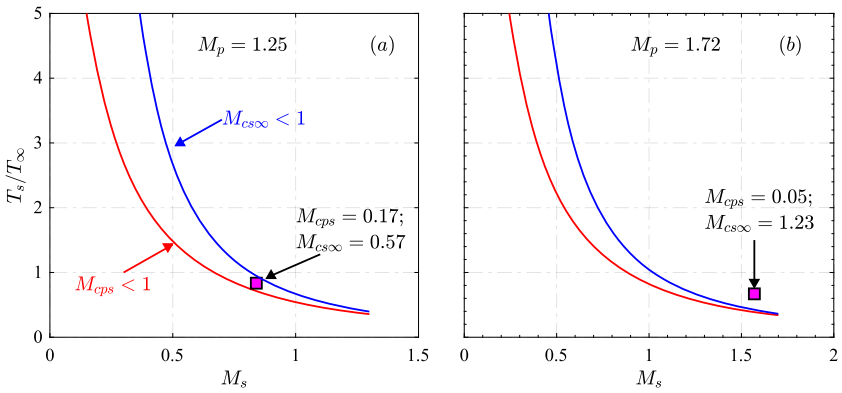


FIG. 5. Mach wave elimination region for a jet: (a) $M_p = 1.25$; (b) $M_p = 1.72$. The secondary flow condition is marked with a square symbol.

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optimum U_s/U_p at which both M_{cps} and $M_{cs\infty} < 1$, leading to complete Mach wave elimination. Evidently, the presence of co-flows in the present investigation for $M_p < 1.42$ eliminates the Mach wave radiation, whereas the Mach wave radiation prevails for $M_p > 1.57$ (Table I).

The two most important characteristics of the jet, namely, the core length and characteristic decay, can be visualized/analyzed from the centerline pressure measurements. To characterize the core length and characteristic decay, we measure the Pitot pressure along the jet centerline up to $25D_p$. The supersonic region contains intense noise sources such as Mach waves and, in the case of imperfectly expanded jets, screech/broadband shock noise.[11] A shorter core length and faster characteristic decay are taken as a measure of large jet mixing and reduced jet noise. Following Kumar and Rathakrishnan,[30] the supersonic core length is estimated from the centerline Pitot pressure measurements and is used to quantify the level of mixing of a free jet with the ambient fluid. The supersonic core length is defined as the distance from the nozzle exit to the axial location along the jet centerline, where the local flow Mach number is equal to 1.0.[11,23]

The questions that arise are as follows: How one can normalize the axial distance to present the supersonic core length and jet spread of a coaxial jet, in a uniform and consistent fashion, so the physical meaning of the coaxial jet is retained? Should one choose D_p or D_s for normalization, unlike D_p for a single jet? On what basis, one may choose the length scale for normalization? Development of scaling laws for coaxial jets is more complicated than that for single jets. Murakami and Papamoschou[10,11] argued that the scaling parameter is not universal and depends on the ultimate goals and applications of the study. That is, the complication for the scaling law arises if the secondary Mach number increases, so do the mass flow rate and momentum of the secondary jet, as in the present investigation. Using the primary flow diameter D_p for normalization means that the comparisons will, in general, be at variable mass and momentum. This is useful in applications where the size of the primary jet is constrained and one wishes to determine the effect of the secondary stream on the mean flow. When there is some freedom in selecting the sizes, shapes, and conditions of the primary and secondary streams, it may be desirable to compare flows at either the constant mass flow rate or constant momentum. That is, for a constant mass flow rate ratio, one may choose D_{eq} for the scaling law as given in Eq. (4). Hence, for comparison, we present in Table II the supersonic core length of the coaxial jet normalized using D_p and D_{eq} along with the core length of the primary jet normalized using D_p. Nevertheless, normalizing by either D_p or D_{eq} captures the elongation of the jet core for the coaxial jet. However, normalizing by D_{eq} results in a marginal increase in L/D_{eq} than L/D_p. Apparently, the increase in L/D_{eq} is due to $\dot{m}_s > \dot{m}_p$. For the coaxial jets reported in the jet literature, it is a usual practice to normalize L_c with D_p.[11,40] For uniformity of the scaling law with those reported in the open literature, we use the D_p for normalizing L_c. In fact, the present study is concerned mainly with the effect of $C2$–$C5$ on jet mixing with respect to $C1$. Interestingly, the $\Delta L^*_{c,C1}$ [Eq. (7)] and $\Delta L^*_{c,C2}$ [Eq. (8)] discussed in Sec. IV A will not vary if L_c is normalized by either D_p or D_{eq}. Thus, in the present study, we use D_p for normalizing L_c, and an asterisk denotes the length scale normalized by D_p. However, it is worth mentioning that there is no detailed study on the scaling law for normalizing the length scale in a coaxial jet, and more attention is required in the future,

$$D_{eq} = D_p\sqrt{(\dot{m}_p + \dot{m}_s)/(\dot{m}_p)}. \tag{4}$$

To examine the L_c^* of $C0$ and $C1$, the variation of p_{0t}/p_{0p} for $C0$ and $C1$ along the jet centerline is shown in Fig. 6. Note that the horizontal dotted line in Fig. 6 represents the cut-off p_{0t}/p_{0p} for the end of the supersonic core length and the vertical dotted line corresponds to L_c^*. The Pitot probe measures the sonic Mach number at $p_{0t}/p_a = 1.89$. Therefore, for $p_{0p}/p_a = 5.0$, the corresponding cut-off p_{0t}/p_{0p} is $1.89/5 \approx 0.37$. Refer to Sec. 3.1 in the study of Kumar and Rathakrishnan[23] for detailed discussion on estimating L_c^* from centerline Pitot pressure measurements. Few important observations can be made. First, the L_c^* of $C0$ and $C1$ is 18 and 24.7, respectively. For $C1$, the presence of secondary flows causes the elongation of the supersonic core length as also observed by several researchers in the open literature.[41,42] Second, the sudden drop in $p_{0t}/p_{0p} = 0.37$ at $x^* = 1.75$ for $C0$ represents the presence of a Mach disk [Fig. 7(a)]. The static pressure behind the Mach disk is approximately equal to the ambient pressure,[52,53] and hence, the pressure downstream of the Mach disk meets the cut-off p_{0t}/p_{0p} for the supersonic core length. The presence of secondary flows for $C1$ retrieves the p_{0t}/p_{0p} drop from 0.37 to 0.55 at $x^* = 1.0$, implying that the Mach disk moves in the upstream direction [Fig. 7(b)]. In fact, the strength of the Mach disk is decreased as seen by the p_{0t}/p_{0p} magnitude. Note that in a supersonic flow, the p_{0t}/p_{0p} decreases as the flow Mach number is increased. Third, the presence of strong waves in the core for $C0$ prevails until $x^* = 12$. The oscillations in p_{0t}/p_{0p} signify the presence of

TABLE II. Non-dimensionalized supersonic core length for M_p and M_s given in Table I.

$C0$	$C1$	
L_c/D_p	L_c/D_p	L_c/D_{eq}
9.2	13.2	16.9
8.4	15.6	19.5
9.2	17.2	20.9
10.9	21.7	25.9
18.7	15.4	30.2
17.9	28.2	33.2

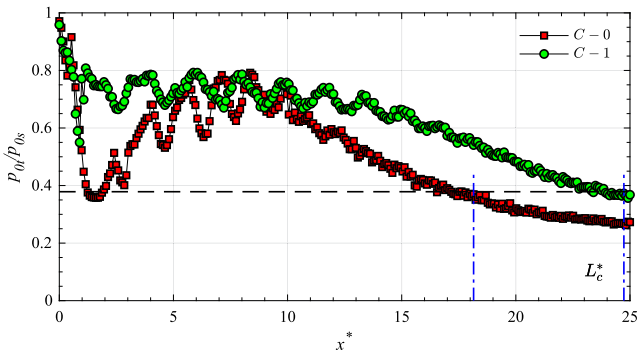


FIG. 6. Centerline Pitot pressure decay for $C0$ and $C1$ at $U_s/U_p = 0.94$.

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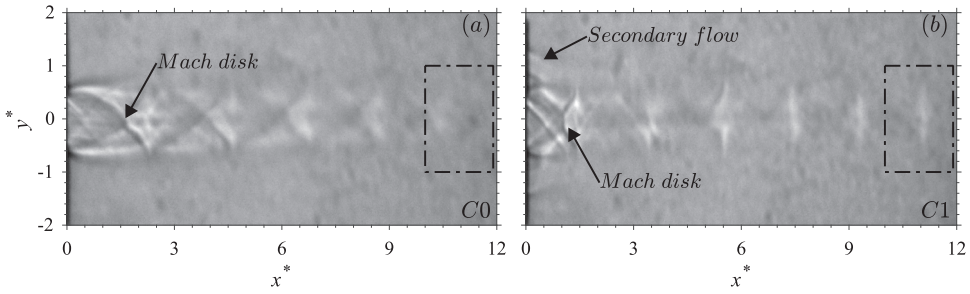


FIG. 7. Shadowgraph images for *C*0 and *C*1 jets along the *xy*-plane at U_s/U_p = 0.94.

strong waves in the jet core. On the other hand, for *C*1, the presence of strong waves in the jet core is seen until $x^* = 16.5$. Shadowgraph images also compliment the presence of strong waves in the core for *C*1 [marked in Fig. 7(b)] than *C*0 [marked in Fig. 7(a)]. That is, the secondary flow shields the primary jet from decaying, and hence, the L_c^* increases.

To envisage the physical reason behind the observed increase in L_c^* for *C*1, we present in Fig. 8 the iso-contours of *C*0 [Figs. 8(a1)–8(a4)] and *C*1 [Figs. 8(b1)–8(b4)] for U_s/U_p = 0.72, 0.85, 0.91, and 0.94. This facilitates to observe the effect of the secondary flow on the development of the primary jet. The zone corresponding to p_{0t}/p_{0p} = 0.8 is substantially elongated due to the presence of secondary

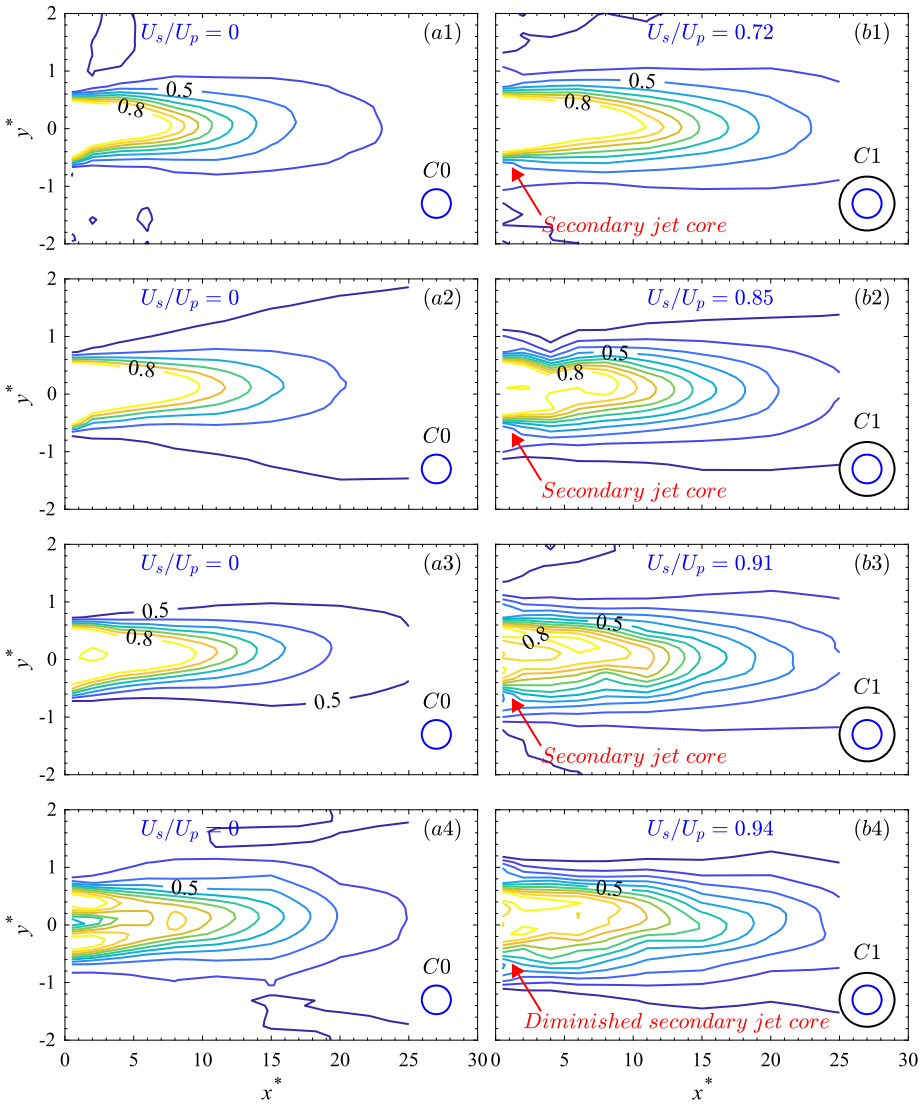


FIG. 8. Iso-Pitot (p_{0t}/p_{0s}) contours for *C*0 (a1–a4) and *C*1 (b1–b4) along the *xy*-plane at various U_s/U_p: (a1 and b1), U_s/U_p = 0.72; (a2 and b2), U_s/U_p = 0.85; (a3 and b3), U_s/U_p = 0.91; (a4 and b4), U_s/U_p = 0.94. Contour interval = 0.05.

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flows (C1) for $U_s/U_p \leq 0.85$ [Figs. 8(b1) and 8(b2)]. The elongated jet stream for C1 [Figs. 8(b1)–8(b4)] than C0 [Figs. 8(a1)–8(a4)] is due to two reasons. First, the increase in p_{0p}/p_a of the primary jet is accompanied by the increase in the primary jet core, as also reported by several researchers.[19,23] Second, as the velocity difference between the primary and secondary streams becomes very small ($U_s/U_p \geq 0.91$), the secondary core is not discernible [Fig. 8(b4)]. In fact, the secondary flow reduces the growth rate of the primary shear layer as seen from the reduced spacing between the contour levels leading to the increased jet core.

The growth rate of the shear layer δ' is measured in the near-field of C0 and C1 using the Pitot pressure profiles as given by Papamoschou and Roshko[42] (see Fig. 5 in Ref. 42) and Murakami and Papamoschou[10] (see Fig. 5 in Ref. 10). In brief, the shear layer grows linearly in the downstream direction. The growth rate of the shear layer was obtained using a linear best-fit of the Pitot thickness in the downstream direction. Evidently, δ' is suppressed by the presence of the secondary flow (Table III). Thus, the strongly coupled behavior of the secondary flow and the primary flow results in the reduced growth of the shear and, hence, the elongation of the jet core. A similar observation was also reported by Papamoschou and Rostamimonjezi[13] who documented the effect of U_s/U_p (=0–1.0) on the characteristics of a Mach 0.9 primary jet. They argued that the primary core length (L_c^*) increases with the increase in U_s/U_p. Interestingly, Papamoschou and Rostamimonjezi[13] correlated that there is a strong similarity between the location of peak noise and the end of the primary core. Thus, the strongest noise comes from the end of the primary core, and the extension of the primary core leads to an elongation of the noise source region.

It would be interesting to compare the growth rate of the shear layer to that predicted using the shear layer model developed by Papamoschou and Roshko.[42] They reported that the normalized growth rates of the coaxial jet shear layer with the incompressible shear layer (δ'/δ'_0) depend on M_{cps}. In fact, the growth rate of the equivalent incompressible shear layer δ'_0 is predicted using Eq. (5) given in Ref. 42, where S is S_{ps} and $S_{p\infty}$ for M_{cps} and $M_{cp\infty}$, respectively. Note that R in Eq. (5) is R_{ps} for M_{cps} and 0 for $M_{cp\infty}$ since the ambient flow velocity $U_\infty \approx 0$. The dependence of δ'/δ'_0 on M_{cps} is shown in Fig. 9, along with the dependence of δ'/δ'_0 on $M_{cp\infty}$. The curve fit [Eq. (6)] of δ'/δ'_0 reported by Murakami and Papamoschou[11] is also shown in Fig. 9. Few interesting observations can be made. First, the δ'/δ'_0 of the present investigation fits reasonably well with those reported in the open literature.[11,43] However,

TABLE III. Growth rates.

C0		C1	
M_{cp_∞}	δ'	M_{cps}	δ'
0.95	0.048	0.17	0.044
1.02	0.038	0.14	0.032
1.10	0.031	0.10	0.023
1.23	0.026	0.07	0.013
1.35	0.022	0.05	0.008
1.45	0.019	0.03	0.006

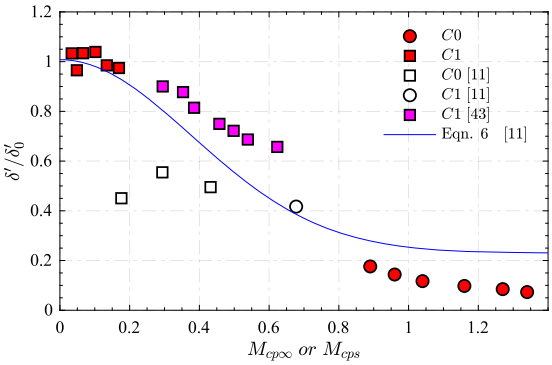


FIG. 9. Dependence of normalized growth rates of coaxial jet shear layers with the convective Mach number M_{cps}.

there is a slight difference with that of the model predicted in Ref. 42. Second, the normalized growth rate of the shear layer decreases as M_{cps} increases. Third, for $M_{cps} > 1$ or $M_{cp\infty} > 1$, the growth rate of the shear layer is only 20% of the incompressible case, revealing the strong compressibility effect at high M_{cps} and/or M_{cps},

$$\delta'_0 = 0.14\frac{(1-R)(1+\sqrt{S})}{(1+R\sqrt{S})}, \tag{5}$$

$$\delta'/\delta'_0 = 0.23 + 0.77e^{-3.5M^2_{cp\infty}}. \tag{6}$$

In summary, for $M_p < 1.42$, the Mack wave is eliminated, and the Mack wave is not eliminated for $M_p > 1.57$ (Table I). The strongly coupled behavior of the secondary flow and the primary flow results in the reduced growth of the shear layer and, hence, the elongation of the supersonic jet core.

IV. MEAN FLOW CHARACTERISTICS OF C2–C5

The presence of co-flows for $M_p < 1.42$ eliminates the Mach wave radiation, whereas the Mach wave radiation prevails for $M_p > 1.57$. Now, one may wonder how effective would be the tab employed at $M_p < 1.42$ and $M_p > 1.57$. That is, whether employing the tabs in the presence of Mach waves is effective in reducing the core length of the jet. As discussed in Sec. III, the primary core is the zone for the source of noise, and the noise can be reduced if the potential core is reduced. Hence, Sec. IV is devoted to characterizing the effect of tab configuration on jet mixing as a measure of supersonic core length. This section is divided into two sections, namely, the jet mixing enhancement (Sec. IV A) and flow structure development (Sec. IV B). The former is devoted to characterizing the effectiveness of jet mixing under manipulation using the Pitot pressure measurements and the latter to understanding the flow structure modifications under manipulation.

A. Jet mixing enhancement

Figure 10 presents the variation of p_{0t}/p_{0p} with x^* for C2–C5 at $U_s/U_p = 0.94$. For reference, the p_{0t}/p_{0s} of C1 is also shown in Fig. 10. In the jet literature, the Pitot pressure oscillations represent the presence of a shock-cell structure in the jet field, and a decrease in

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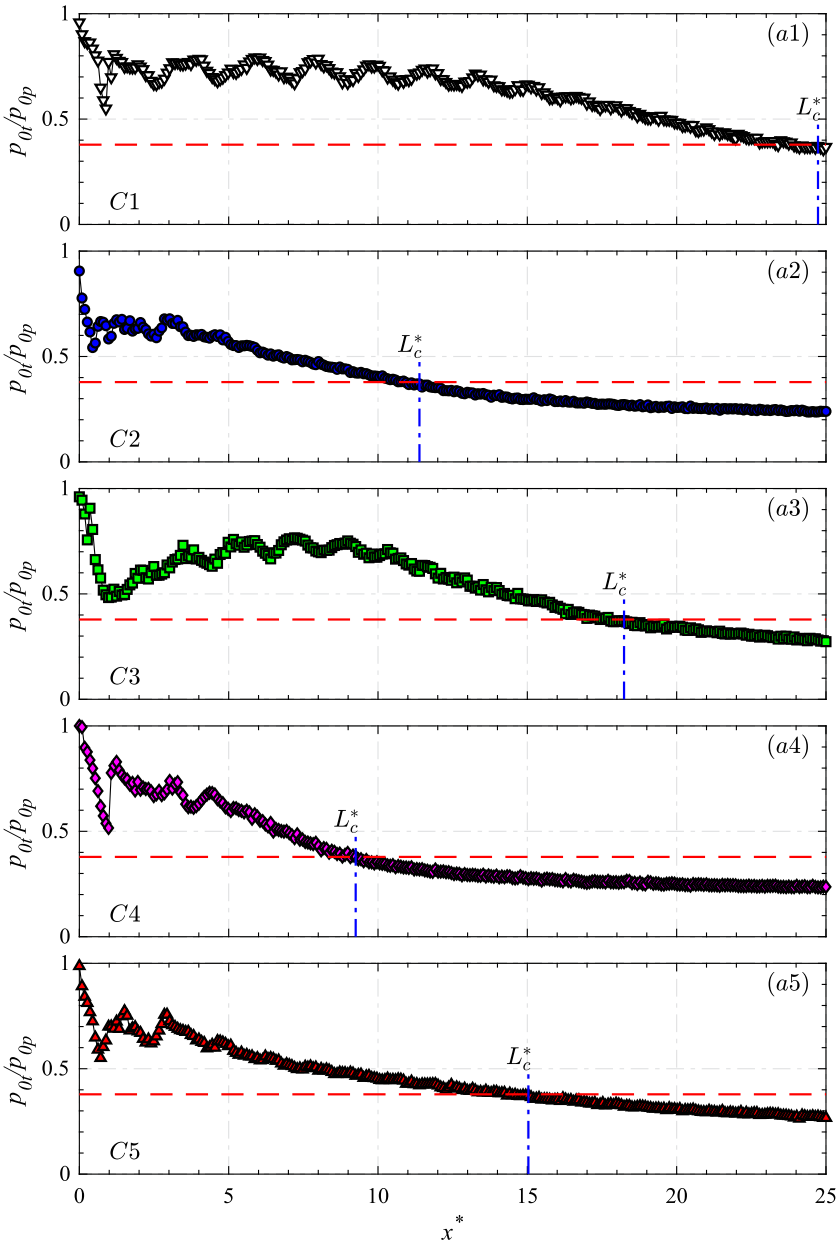


FIG. 10. Centerline Pitot pressure decay for the manipulated jet at U_s/U_p = 0.94. (a1) Coaxial jet (*C*1). (a2) Coaxial jet with the tab in a primary flow (*C*2). (a3) Coaxial jet with the tab in a secondary flow (*C*3). (a4) Coaxial jet with the tab in primary and secondary flows with 0° orientation (*C*4). (a5) Coaxial jet with the tab in primary and secondary flows with 90° orientation (*C*5).

the Pitot pressure can be taken as the decreased shock-cell strength of the jet.[23] In addition, the distance between two peaks represents the length of the shock cell. Few important observations can be made with respect to Fig. 10. First, *C*1 exhibits the strong shock-cell nature of the coaxial jet as explicitly seen from the pressure oscillations [Fig. 10(a1)]. The L_c^* of *C*1 is about 24.7. The Pitot pressure drops immediately downstream of the nozzle exit at $x^* = 0.9$. This drop is caused by the Mach disk as seen in the shadowgraph images [Fig. 10(a1)], which will be discussed in Sec. IV B. Downstream of the Mach disk, the Pitot pressure recovers considerably although it undergoes several oscillations during the recovery. The recovery is due to the transverse transfer of momentum across the slip stream from the outer region to the inner region. The oscillations are caused due to the expansion and compression waves encountered by the centerline flow as it moves through the shock cells. Second, for *C*2 [Fig. 10(a2)], there is a dip in p_{0t}/p_{0p} just at the nozzle exit, lowering the value of p_{0t}/p_{0s} to 0.55, and the Pitot pressure recovery is only marginal. This is due to the deflection of the primary jet away from the centerline by the primary tabs. When the tabs are introduced, the shock envelope in front of the tabs results in significant reduction in Pitot pressure and hence the reduced p_{0t}/p_{0s} compared with that for *C*1. The p_{0t}/p_{0p} then recovers slightly as the high stagnation pressure fluid moves toward the centerline. The p_{0t}/p_{0p} then decays gradually, which may be because of the shear layer reaching the centerline

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and causing the mixing of the fluid in the central region. The L_c^* of $C2$ is about 11.4, which is about 54% reduction in the core length ($\Delta L_{c,C1}^*$) than that in $C1$. The shock-cell structure is made weaker by the presence of the tab as seen explicitly from the reduced Pitot pressure. Third, for $C3$ [Fig. 10(a3)], the p_{0t}/p_{0p} recovers at a faster rate and exhibits a larger p_{0t}/p_{0s} than that for $C2$. Evidently, the strength of the shock cell and number of shock cells are increased in comparison with those in $C2$, leading to reduced mixing and hence a larger L_c^* of 18.2 [Fig. 10(a3)]. It should also be noted that the core length reduction achieved by $C3$ is only 25%. Fourth, the L_c^* of $C4$ is about 9.3 and is the least among the configurations examined in this study [Fig. 10(a4)]. It should also be noted that the core length reduction achieved for $C4$ is the highest (61%), revealing the best tab configuration for jet mixing. The characteristic decay for $C4$ is found to be faster than that for $C2$ and $C3$. However in the jet field beyond $x^* = 15$, the p_{0t}/p_{0p} is almost identical for both $C2$ and $C4$. The shock-cell structure is weaker for $C4$ but not as weak as that observed for $C2$. Fifth, the $\Delta L_{c,C1}^*$ of $C5$ [Fig. 10(a5)] is about 40%, which is significantly smaller than that for $C2$ (=54%) and $C4$ (=61%). Note that for $C5$, $L_c^* = 15$. Last, both the reduction in the core length and weakening of shock-cell structures are favorable from aerodynamic mixing and aero-acoustic points of view. The former can be regarded as reduction of the infrared source zone, and the latter can be taken as an indication of elimination of broadband shock noise.[29]

From the preceding discussions, it is well evident that L_c^* is strongly influenced by the tab configuration. Now, one may wonder how L_c^* would vary with U_s/U_p. Figure 11(a) presents the variation of L_c^* with U_s/U_p. The L_c^* of $C1$ is also shown for comparison. The L_c^* of $C1$ monotonically increases with the increase in U_s/U_p. Once manipulated, the L_c^* of $C2$–$C5$ is substantially reduced. As such, the L_c^* of $C2$–$C5$ also increases with the increase in U_s/U_p. Evidently, the L_c^* of $C3$ is the largest among the manipulated jets, implying the least mixing promoting efficiency, at all U_s/U_p of the present investigation. For all the jets, L_c^* increases slowly for $U_s/U_p \leq 0.84$ but increases rapidly for $U_s/U_p \geq 0.91$. Indeed, the L_c^* of $C2$, $C4$, and $C5$ is almost the same for $U_s/U_p \leq< 0.85$, whereas for $U_s/U_p \geq 0.91$, the L_c^* of $C4$ is the least of the manipulated jets, highlighting a better controlling technique for jet mixing. In other words, the performance of $C4$ is superior to that of other configurations at all levels of expansion.

As L_c^* of $C1$ increases with the increase in U_s/U_p, it would be interesting to interpret jet mixing of the manipulated jet with respect to $C1$. Hence, we introduce the percentage reduction in the core length with respect to $C1$ ($\Delta L_{c,C1}^*$) to quantify the mixing effectiveness of the manipulated jet,

$$\Delta L_{c,C1}^* = \left(1 - \frac{L_{c\ Manipulated\ jet}^*}{L_{c\ ,C1}^*}\right) * 100. \tag{7}$$

Figure 11(b) shows the variation of $\Delta L_{c,C1}^*$ with U_s/U_p for $C2$–$C5$. As evident from Fig. 11(b), $\Delta L_{c,C1}^*$ for $C2$, $C4$, and $C5$ decreases with the increase in U_s/U_p. The reduced $\Delta L_{c,C1}^*$ for $U_s/U_p > 0.85$ is primarily due to two reasons. First, for $U_s/U_p > 0.85$, the Mach number of the secondary jet is supersonic ($M_s > 1$). Papamoschou and Rostamimonjezi[13] documented that as the secondary flow reached sonic conditions, the spreading rate was significantly enhanced. In contrast, increasing the secondary flow to supersonic velocity decreased the spreading rate. Note that in the present study, the supersonic secondary flow is achieved for $U_s/U_p > 0.85$. Second, for $U_s/U_p > 0.85$, the Mach wave radiation occurs for the secondary jet as evident by $M_{cs\infty} > 1$. Both the scenarios cause the $\Delta L_{c,C1}^*$ to reduce for $U_s/U_p > 0.85$. Interestingly, the $\Delta L_{c,C1}^*$ is the largest of the manipulated jets in the absence of Mach wave radiation. Presumably, the jet mixing for $U_s/U_p > 1$ might increase if $M_{cs\infty}$ is reduced. Notably, $\Delta L_{c,C1}^*$ of $C3$ is almost invariant

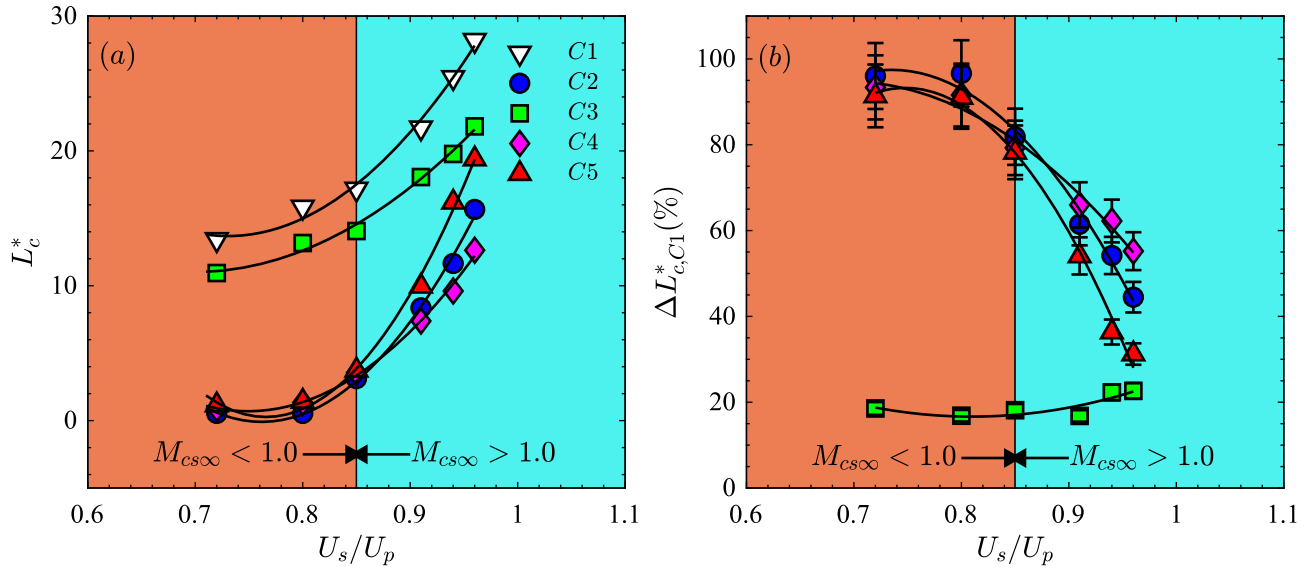


FIG. 11. (a) Dependence of L_c^* on U_s/U_p for the manipulated jets $C1$–$C5$. (b) Dependence of $\Delta L_{c,C1}^*$ on U_s/U_p for the manipulated jets $C2$–$C5$ with respect to $C1$. The $M_{cs\infty}$ corresponds to the respective U_s/U_p as in Table I and Fig. 5.

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with U_s/U_p. This might be due to the sole effect of secondary tabs impacting the supersonic jet arising from the secondary nozzle at high p_{0s}/p_{0p} (Fig. 2) or p_{0s}/p_a (Table I). The largest $\Delta L^*_{c,C1}$ is achieved for $C2$ followed by $C4$ and $C5$, and the least jet mixing is observed for $C3$. Indeed, it should be noted that for $C2$ and $C3$, the jet is manipulated by two tabs. Since the secondary tab for $C4$ is aligned 0° with the primary tab, $C4$ is analogous to two tab configurations. Bradbury and Kadhem[20] and Samimy *et al.*[33] documented that when the number of tabs was reduced to 2 from 4, the apparent potential-core length was reduced followed by a rapid decay of the centerline mean velocity. Hence, the $\Delta L^*_{c,C1}$ is the largest for $C4$ than $C5$.

Murakami and Papamoschou[10] investigated a Mach 1.5 primary jet surrounded by a Mach 0.9 secondary jet using an eccentric jet with exactly the same flow parameter as the coaxial jet. They documented that L^*_c of an eccentric configuration was about 13.0, in comparison with 17.4 for the co-axial configuration. That is, a core length reduction $\Delta L^*_{c,C1}$ of about 26% was achieved with the eccentric configuration. The present investigation at the similar flow condition (M_p = 1.42 and M_s = 1.15) for $C2$, $C4$, and $C5$ achieves a $\Delta L^*_{c,C1}$ of about 61%, 66%, and 55%, respectively. It should be noted that the primary and secondary flow Mach numbers in the present investigation are marginally lower than those in Ref. 10. Nevertheless, the core length reduction of the tabs is significantly larger than that of an eccentric configuration reported in Ref. 10. On the other hand, it should be noted that there is a significant thrust loss associated with the tab, in contrast to no thrust loss in an eccentric configuration.

Now, one may wonder how effective is the jet manipulated by tabs in comparison with $C0$. Here, we compare the core length reduction of the manipulated co-flow jet (M_s = 1.15) with that of the primary jet at M_p = 1.42. The core length reduction of 23% is achieved for $C2$ (L^*_c = 8.3) in comparison with $C0$ (L^*_c = 10.9). In contrast, under the identical flow conditions, the eccentric configuration results in an elongation of the core length by 15%. Note that the L^*_c of the primary jet and eccentric co-flow jets is around 11.3 and 13.0, respectively.[10] Evidently, the jet manipulated by tabs is superior to the eccentric configuration.

It is worth noting that the tab blockage for $C4$ and $C5$ is 10%, which is twice that of $C2$ and $C3$ (=5%). Now, one may question what additional jet mixing benefits are accompanied by introducing the secondary tabs on $C2$, the $C4$ configuration (the net tab blockage is 10%). We introduce the percentage reduction in the core length ($\Delta L^*_{c,C2}$) to quantify the mixing effectiveness of the manipulated jet ($C4$ and $C5$) with respect to $C2$,

$$\Delta L^*_{c,C2} = \left(1 - \frac{L^*_{c\ Manipulated\ jet}}{L^*_{c,C2}}\right) * 100. \tag{8}$$

Since the least jet mixing is observed for $C5$, as expected, the $\Delta L^*_{c,C2}$ exhibits a negative value for $C5$ at all U_s/U_p. On the other hand, $C4$ increases the jet mixing only by 20% for $U_s/U_p \geq 0.91$ (Fig. 12). It is worth noting that the 20% increase in jet mixing associated with $C4$ is due to the increase in tab blockage to 10%. Now, one would wonder whether the increased jet mixing for $C4$ is due to the increased tab blockage or the effect of the tab in the secondary jet. Interestingly, Zaman[22] documented that the jet mixing increases with the increase in tab blockage, and hence, the higher jet mixing for $C4$ may be due to the increase in tab blockage. In line with the same observation,

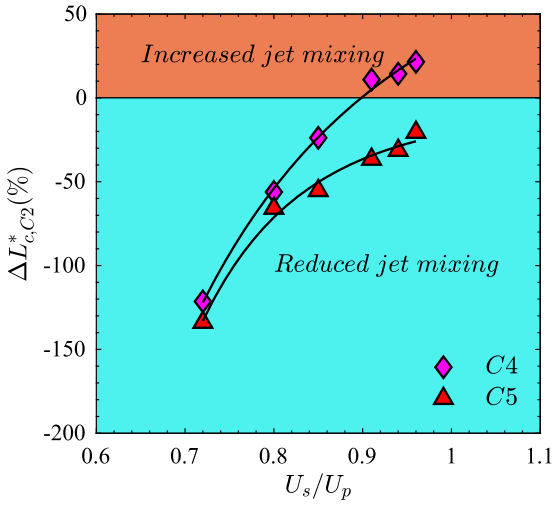


FIG. 12. Dependence of $\Delta L^*_{c,C2}$ on U_s/U_p for the manipulated jets $C4$ and $C5$ with respect to $C2$.

the jet mixing decreases for $C5$ even with the increase in tab blockage. Hence, the increase in jet mixing for $C4$ is not only due to the increase in tab blockage but also due to the orientation of the tab in the secondary jet.

In summary, the $\Delta L^*_{c,C1}$ of $C2$ and $C4$ is the highest among the manipulated jets, highlighting the better controlling technique for jet mixing. However, the former introduces a tab blockage of 5% and the latter 10%. In view of the additional 5% blockage introduced by $C4$, the increase in $\Delta L^*_{c,C1}$ achieved by $C4$ is only a mere 20%. In contrast, $C3$ and $C5$ exhibit the least jet mixing of the tab investigated in the present study. Hence, tabs in the secondary jet are not beneficial from the aspect of jet mixing.

B. Flow structure development

This section addresses the physical mechanism behind the jet manipulation based on detailed pressure measurement and shadowgraph images captured along the jet stream. To get a physical insight into the physics behind the jet mixing for the manipulated jet at U_s/U_p = 0.94 (Fig. 11), we present in Figs. 13 and 14 the Pitot pressure measured along the xy- and xz-planes, respectively. The streamwise vortices generated from the tab and their associated circulation dictate the jet mixing capability and hence the jet spread.[46] In fact, it would be interesting to see the effect of tabs on jet spread as a measure of jet mixing.

Few important observations can be made. First, the jet spreads faster in the y direction [Figs. 13(a1)–13(a4)] than the z direction [Figs. 14(a1)–14(a4)]. The larger the jet spread, the higher the jet mixing. Second, for $C2$, $C4$, and $C5$, the two pressure peaks on either side of the jet axis along the y direction [Fig. 13(a1)] reveal that the tabs split the jet into two streams, which is taken as an indication of jet bifurcation. The bifurcation for $C2$, $C4$, and $C5$ prevails even until x^* = 11.0 [Fig. 13(a4)]. We also note that for $C1$ and $C3$, the two pressure peaks in Fig. 14(a1) are an indication of the shock-cell structure and not the jet bifurcation. That is, the Pitot pressure increases as we move outward from the jet

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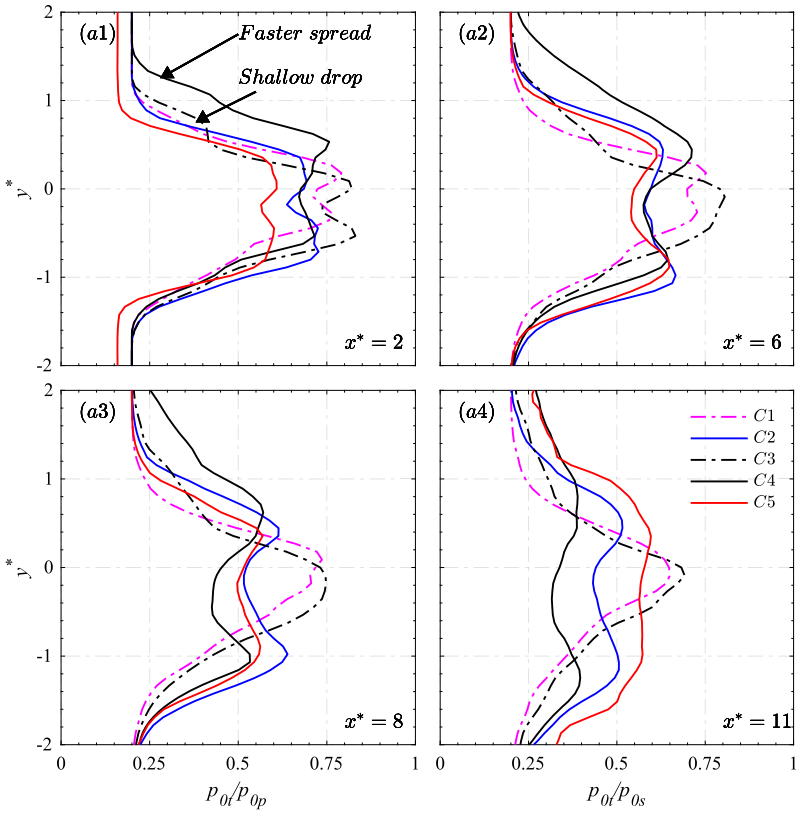


FIG. 13. Pressure profiles at various axial locations of the manipulated jets for U_s/U_p = 0.94 along the *xy*-plane: (a1) x^* = 2; (a2) x^* = 6; (a3) x^* = 8; (a4) x^* = 11. *C*1, natural coaxial jet; *C*2, coaxial jet with the tab in a primary flow; *C*3, coaxial jet with the tab in a secondary flow; *C*4, coaxial jet with the tab in primary and secondary flows with 0° orientation; *C*5, coaxial jet with the tab in primary and secondary flows with 90° orientation.

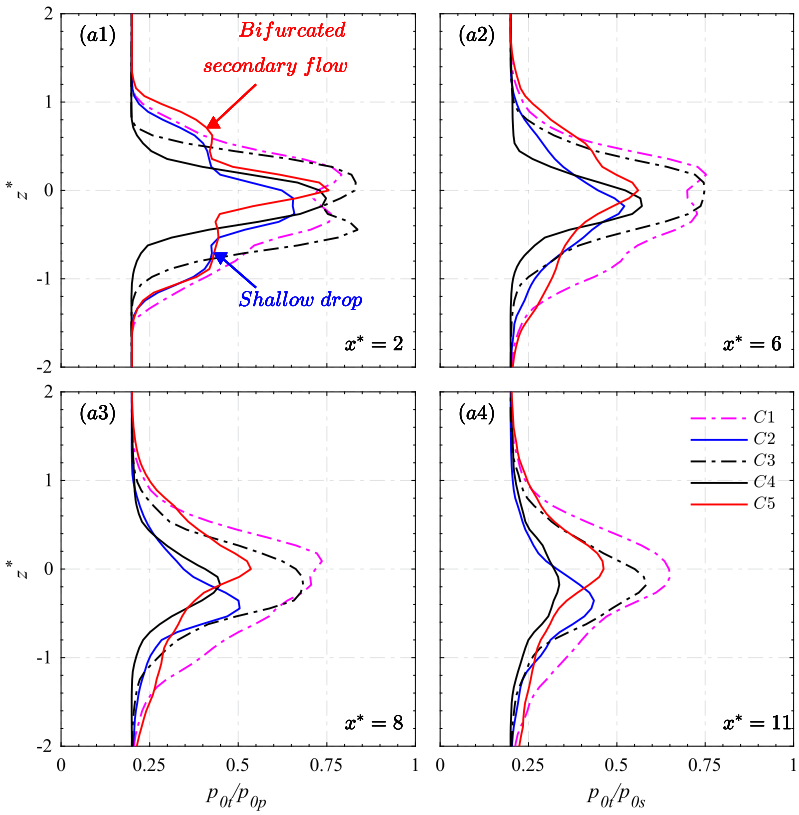


FIG. 14. Pressure profiles at various axial locations of the manipulated jets for U_s/U_p = 0.94 along the *xz*-plane: (a1) x^* = 2; (a2) x^* = 6; (a3) x^* = 8; (a4) x^* = 11. *C*1, natural coaxial jet; *C*2, coaxial jet with the tab in a primary flow; *C*3, coaxial jet with the tab in a secondary flow; *C*4, coaxial jet with the tab in primary and secondary flows with 0° orientation; *C*5, coaxial jet with the tab in primary and secondary flows with 90° orientation.

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axis until the shock boundary interferes reaching a maximum, and then the Pitot pressure decreases further. It is worth mentioning that the increase in Pitot pressure for a supersonic flow implies the Mach number is decreasing.[23] Hence, the Mach number decreases from the jet axis as we move toward the shock boundary, and thereafter, the Mach number again decreases in subsonic flows as the Pitot pressure decreases. Notably, the two peaks observed in Figs. 14(a2)–14(a4) for *C*1 and *C*3 are not an indication of jet bifurcation. In other words, the p_{0t}/p_{0p} for *C*1 and *C*3 exhibits a single core [Figs. 13(a3), 13(a4), 14(a3), and 14(a4)] and not the jet bifurcation. On the other hand, the tabs converge the jets in the *z* direction [Figs. 14(a1)–14(a4)]. Third, for *C*2, the jet exhibits a shallow drop in p_{0t}/p_{0p} at $x^* = 2.0$ [Fig. 14(a1)], revealing the undisturbed secondary flow due to the absence of secondary tabs. Fourth, *C*4 exhibits a faster spread in the *y* direction [marked in Fig. 13(a1)] and a relatively reduced spread in the *z* direction [Fig. 14(a1)]. The differential jet spread in the *y* and *z* directions is a characteristic of an elliptic jet[44,45] and may lead to axis-switching of the jets.[46] Axis switching is a phenomenon in which the cross section of an asymmetric jet evolves in such a manner that after a certain distance from the nozzle, the major and the minor axes are interchanged.[46] Even though the axis-switching phenomenon was not observed in the present investigation, the tab employed on a circular jet (*C*4) might behave as a non-circular jet emulating the characteristics of an elliptic jet. The tab makes the uniform azimuthal radius of curvature at the nozzle exit become asymmetric. As such, the flow along the *z* direction and normal to the tab (*y* direction) emulates the minor and major axes of an elliptic jet, respectively. Similarly, Seidel *et al.*[47] carried out an experimental investigation on the effects of various radial blowing configurations on the decay and mixing of a circular main jet. Their results revealed that the mixing performance of noncircular jets could be emulated via carefully arranging steady control jets around a circular main jet. It is worth mentioning that the shallow drop in p_{0t}/p_{0p} at $x^* = 2.0$ [Fig. 14(a1)] is absent for *C*4 as a result of the flow disturbed by secondary tabs. In *z*-profiles, p_{0t}/p_{0p} is narrower at all axial distances compared with that in the *y*-profiles. This is because in the *z* direction, the flow could spread greatly because of the absence of any solid body. Fourth, for *C*5, the jet exhibits a

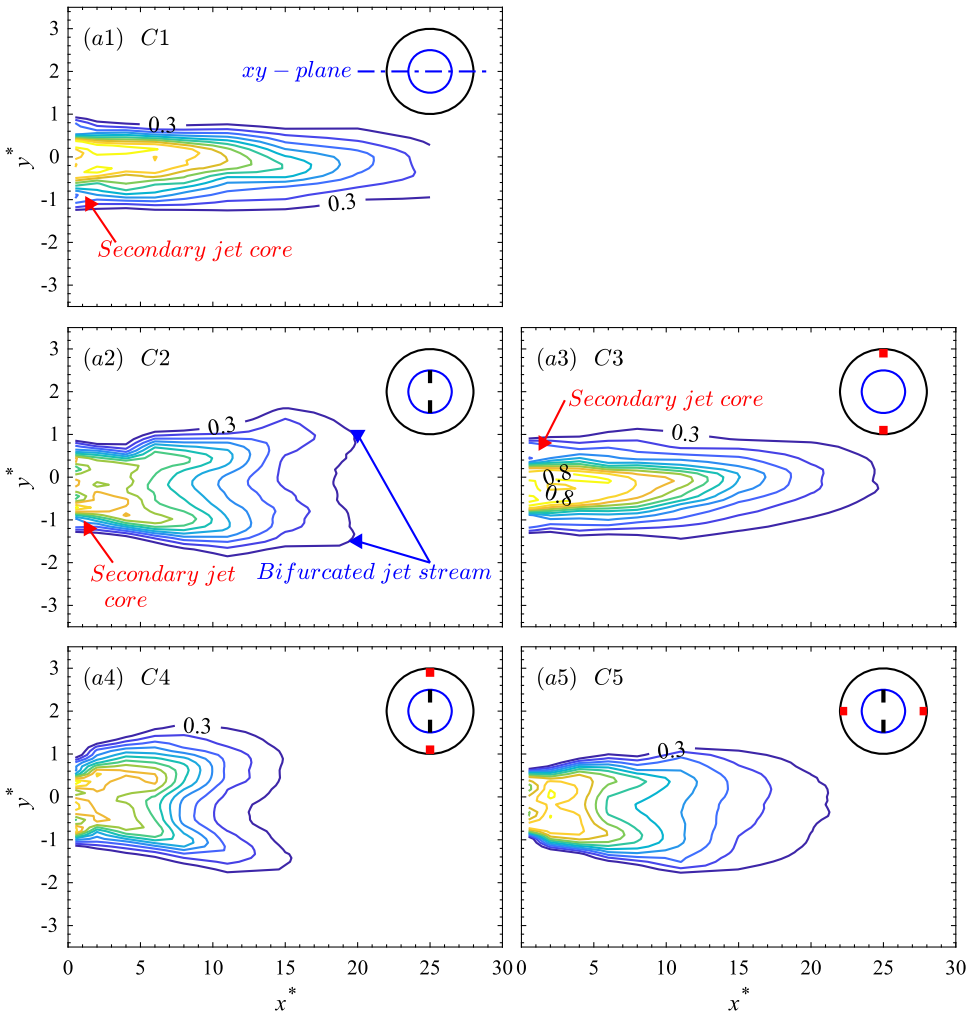


FIG. 15. Iso-Pitot (p_{0t}/p_{0s}) contours of the manipulated jets along the *xy*-plane at $U_s/U_p = 0.94$. The dotted line in the inset figure represents the measurement plane. Contour interval = 0.05. (a1) Coaxial jet (*C*1). (a2) Coaxial jet with the tab in a primary flow (*C*2). (a3) Coaxial jet with the tab in a secondary flow (*C*3). (a4) Coaxial jet with the tab in primary and secondary flows with 0° orientation (*C*4). (a5) Coaxial jet with the tab in primary and secondary flows with 90° orientation (*C*5).

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minor peak about ± z = 0.8 at x^* = 2.0 [Fig. 14(a1)], revealing the jet is bifurcated due to the secondary tabs. Evidently for $C5$, the jet is bifurcated in both the y and z planes of the manipulated jet. Note that for $C5$, the primary tabs and secondary tabs were employed along the y and z planes, respectively. Evidently, the primary tabs and secondary tabs bifurcate the primary and secondary jets in the y and z planes, respectively. Interestingly, for $C5$, the bifurcating jet on either side of the jet axis coalesces to form a single jet [Fig. 13(a4)], which is not evident for $C2$ and $C4$. This results in the reduced jet spread and hence reduced jet mixing for $C5$. Last, the pressure profiles at x_t^* = 8 [Figs. 13(a3) and 14(a3)] and 11 [Figs. 13(a4) and 14(a4)] are minimum for $C4$. This is a clear indication of increased momentum transport caused by $C4$.

To study the near-field and far-field mixing caused by the manipulated jets, we present in Figs. 15 and 16 the iso-Pitot pressure contours in the xy- and xz-planes, respectively. We also present here the typical shadowgraph images captured along the xy- and xz-directions of the jet in Fig. 17. Shadowgraph images of $C1$ are also shown for comparison [Fig. 17(a1)]. The growth of the jet along the y and z directions of $C1$ is suppressed due to the presence of the secondary jet. As such, the core of the secondary jet is explicitly seen in Fig. 15(a1). The secondary core protects the primary core from decaying.[13] Note that U_s/U_p = 0.94 corresponds to p_e/p_a = 1.57, the under-expanded state of a Mach 1.4 jet. Hence, a Mach disk is seen immediately downstream of the nozzle exit [Fig. 17(a1)], which was also explicitly seen by the reduced Pitot pressure in Fig. 10(a1).

In general, the tabs distort the shock-cell structure of the jet and cause strong instability waves that propagate both upstream and downstream [Figs. 17(a1)–17(a5)]. As expected, the iso-Pitot contours for $C2$, $C4$, and $C5$ show the sign of jet bifurcation in the xy-plane [Figs. 15(a2), 15(a4), and 15(a5)]. This observation is also complimented by the shadowgraph images in the xy-plane [Figs. 17(a2), 17(a4), and 17(a5)]. The shadowgraph images clearly demonstrate the difference in the wave domination in the supersonic core and the effect of additional or augmented mixing of $C2$, $C3$, $C4$, and $C5$ [Figs. 17(a2)–17(b5)] compared with $C1$ [Figs. 17(a1) and 17(b1)]. It is well known that shock-cell length reduction and weakening of waves in the jet core can be taken as a measure of reduced

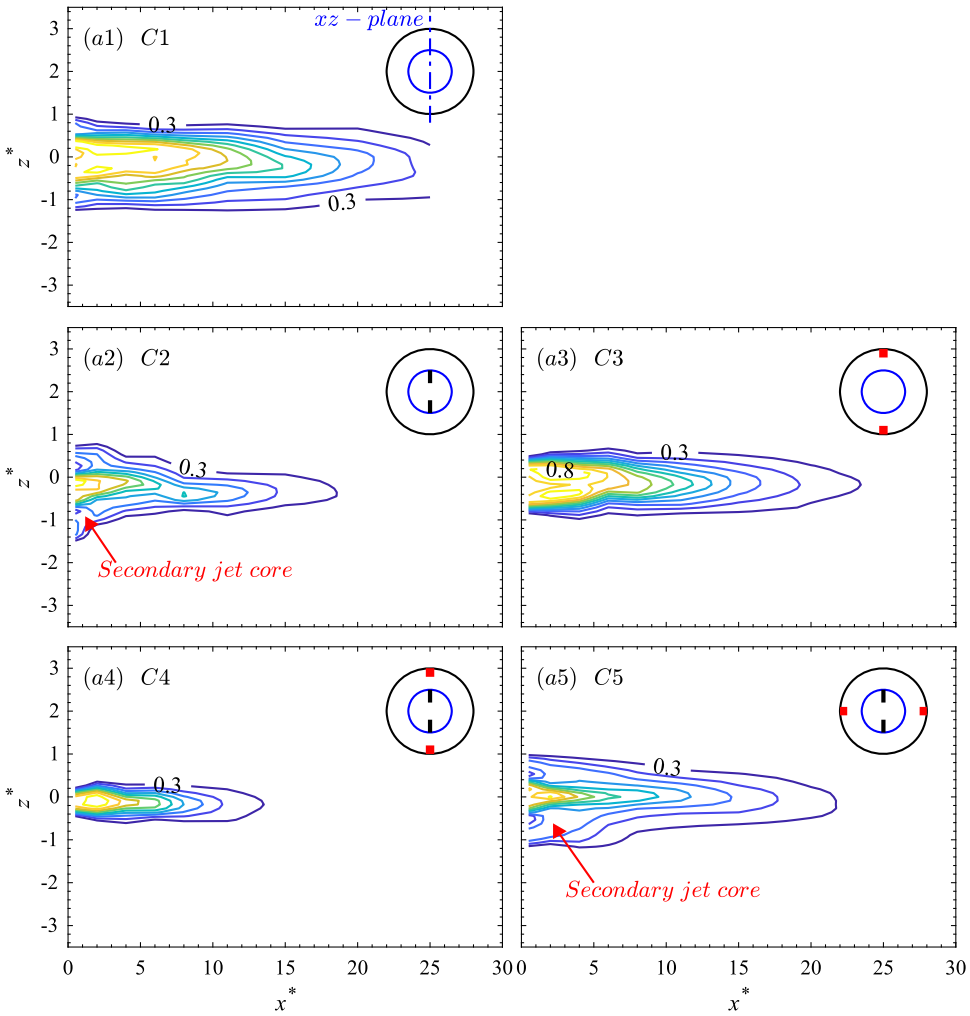


FIG. 16. Iso-Pitot (p_{0t}/p_{0s}) contours of the manipulated jets along the xz-plane at U_s/U_p = 0.94. The dotted line in the inset figure represents the measurement plane. Contour interval = 0.05. (a1) Coaxial jet ($C1$). (a2) Coaxial jet with the tab in a primary flow ($C2$). (a3) Coaxial jet with the tab in a secondary flow ($C3$). (a4) Coaxial jet with the tab in primary and secondary flows with 0° orientation ($C4$). (a5) Coaxial jet with the tab in primary and secondary flows with 90° orientation ($C5$).

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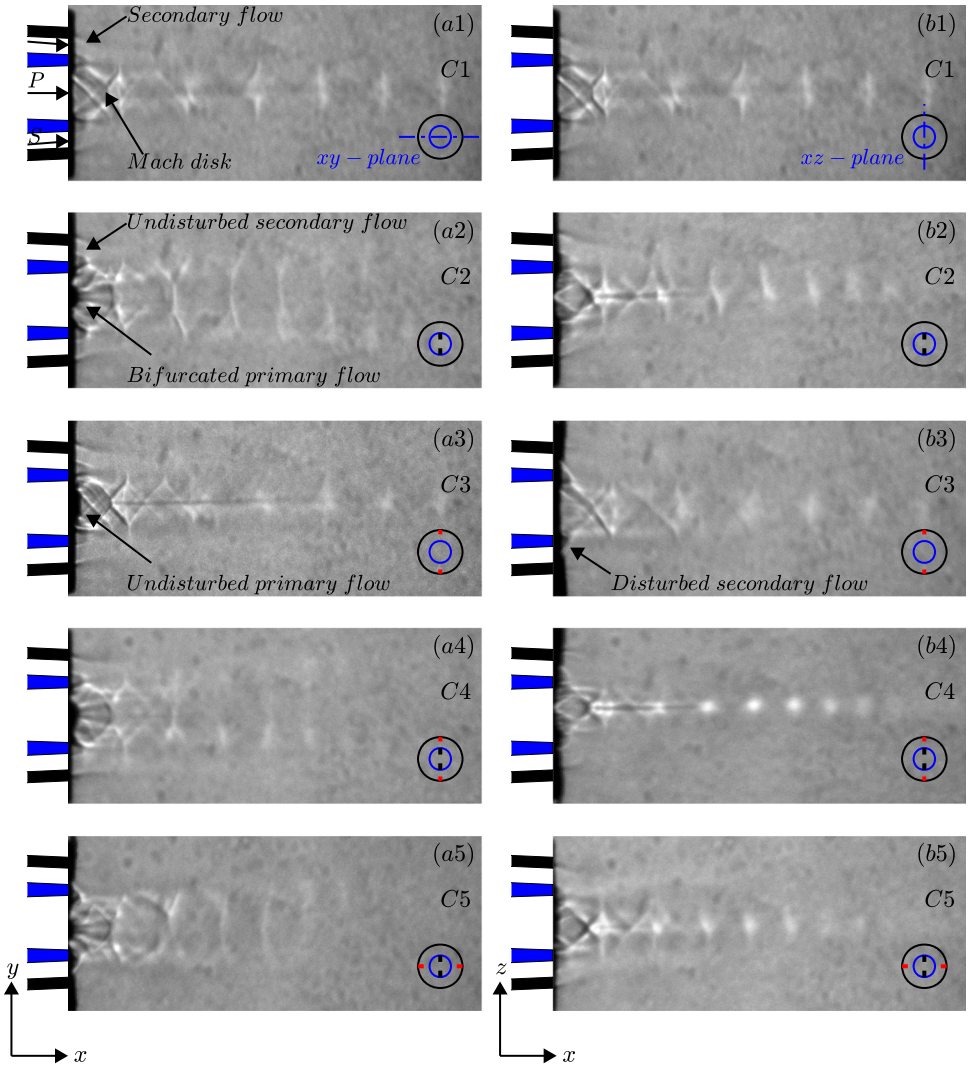


FIG. 17. Shadowgraph images for the manipulated jet at U_s/U_p = 0.94 along the *xy*- and *xz*-planes. (a1) and (b1) Coaxial jet (*C*1). (a2) and (b2) Coaxial jet with the tab in a primary flow (*C*2). (a3) and (b3) Coaxial jet with the tab in a secondary flow (*C*3). (a4) and (b4) Coaxial jet with the tab in primary and secondary flows with 0° orientation (*C*4). (a5) and (b5) Coaxial jet with the tab in primary and secondary flows with 90° orientation (*C*5). The dotted line in the inset figure represents the plane of visualization.

shock associated noise.[29] Therefore, it can be justifiably stated that even though jet noise has not been measured, the manipulated jet is advantageous from the jet noise reduction point of view also. For *C*2, the jet spreads faster in the *xy*-direction [Fig. 15(a2)] than the *xz*-direction [Fig. 16(a2)]. Such an increase in the jet spread in the *xy*-direction would be expected due to an increase in the mass entrainment of the ambient fluid in the *xz*-plane. Hence, the jet spread decreases in the *xz*-direction [Fig. 16(b2)]. A similar observation is also reported by Yang and Zhou[48] who manipulated a turbulent axisymmetric jet using two radial injectors and documented that a strong entrainment of the ambient fluid into the jet was observed along the *xz*-plane and a very strong outward spread of the fluid from the jet in the *xy*-plane. Evidently, the tab employed on a circular jet (*C*4) might behave as a non-circular jet emulating the characteristics of an elliptic jet. One can now see that the core of the secondary jet diminished or eliminated in both the *xy* [Fig. 15(b2)] and *xz* [Fig. 16(b2)] planes due to the presence of the tabs leading to the faster decay of the primary jet. Hence, the strong coupled nature of the jet bifurcation along with the elimination of the secondary jet core results in a faster jet spread leading to better jet mixing. On the contrary, for *C*2, the bifurcation of the coaxial jet is absent for *C*3 [Figs. 15(a3) and 16(a3)] and also reveals a strong core for the secondary flow [Fig. 15(a3)]. Interestingly, the primary jet exhibits an undisturbed flow in both the *xy*- and *xz*-planes as seen in Figs. 17(a3) and 17(b3). Presumably, now one can infer that the reduced jet spread is primarily due to the absence of jet bifurcation and strong core of the secondary jet. The jet spreads at a faster rate for *C*4 [Fig. 15(a4)] than *C*2 [Fig. 16(b4)]. The p_{0t}/p_{0s} of 0.3 occurs

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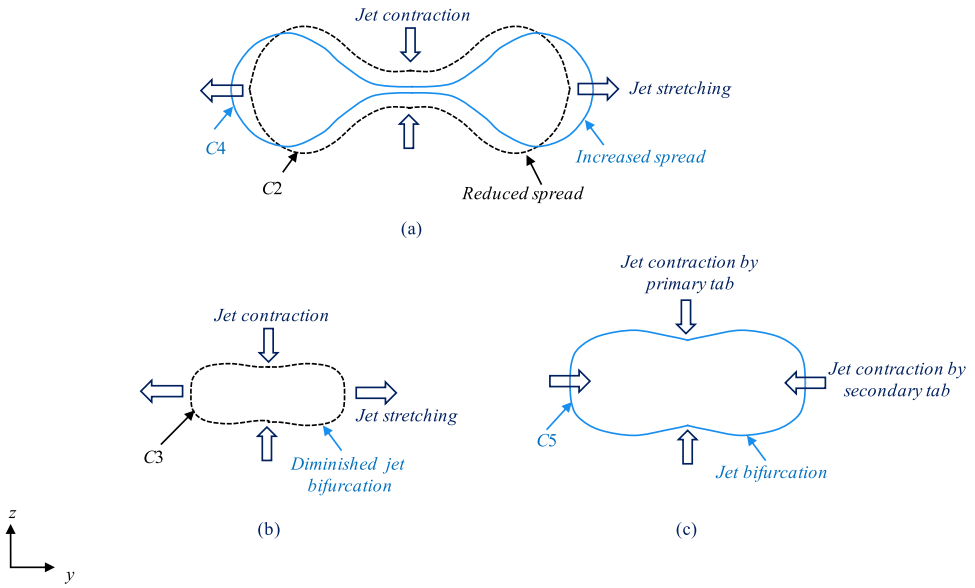


FIG. 18. Proposed sketch of the flow development in the *yz*-plane under different manipulation: (a) *C*2 and *C*4; (b) *C*3; (c) *C*5.

at $x^* = 20$ [Fig. 15(a2)] and 15 [Fig. 15(a4)] for *C*2 and *C*4, respectively. Hence, *C*4 decays earlier than *C*2. The flow characteristic of *C*4 is similar to that of *C*2, where the secondary tabs accelerate the mixing for *C*4. On the contrary, for *C*5, the secondary tabs decelerate the mixing even less than those of *C*2 [Figs. 11(b) and 12]. For *C*5, the bifurcated jet coalesces to form a single jet at the earliest as seen in Fig. 15(a5). Additionally, the core of the secondary jet is stronger as evident in Fig. 16(a5). In other words, the secondary tabs for *C*5 shield the primary jet from decaying, and this results in the reduced jet spread and hence reduced jet mixing.

Based on experimental results, we present in Fig. 18 the schematic of the flow development in the *yz*-plane for *C*2–*C*5. For *C*2, the tab in the primary flow bifurcates the jet in the *xy*-plane [Fig. 15(a2)] and the jet shrinks in the *xz*-plane [Fig. 16(a2)]. Similar to *C*2, *C*4 also exhibits the jet bifurcation in the *xy*-plane [Fig. 15(a4)] and shrinks in the *xz*-plane [Fig. 16(a4)]. However, the jet bifurcation is stronger for *C*4 than *C*2. It is worth noting that the secondary tabs aligned at 0° with the primary tabs for *C*4 result in the increased jet bifurcation leading to the stretching of the jet in the *xy*-plane [Fig. 18(a)]. On the other hand, the jet contracts in the *xz*-plane [Fig. 18(a)]. Notably, the rate of jet shrinking in the *xz*-plane is also faster for *C*4 than *C*2. As we observe, the jet spreads at a faster rate in the *xy*-plane for *C*4 than *C*2, marked as increased and reduced spread, respectively, in Fig. 18(a). The differential spread along the *xy*- and *xz*-planes *C*2 and *C*4 emulates the characteristics of a non-circular jet. Evidently, this increased jet spread for *C*4 results in a faster decay of the jet core and, hence, a higher jet mixing. On the contrary, for *C*3, the tabs in the secondary flow also cause the jet to become asymmetric [Figs. 15(a3) and 16(a3)] as presented in Fig. 18(b). Unexpectedly, the jet bifurcation is negligible for *C*3. As one would expect for *C*5, the tabs in the secondary flow that is aligned 90° with the primary tabs shield the jet to spread in the *xy*-plane. Similarly, the tabs in the primary jet act as an obstacle in shrinking the jet in the *xz*-plane. That is, the jet stretching is eventually halted by the secondary tabs and leads to jet contraction in the *xy*-plane [Fig. 18(c)]. Hence, the jet spread decreases, resulting in reduced mixing for *C*5 than *C*4. As documented by Samimy *et al.*,[33] *C*5 may regain its axisymmetric shape earlier than *C*4. Manipulation of a Mach 1.63 jet by two and four tabs reveals that the jet with the two and four tabs regains the axisymmetric shape by $x^* = 30$ and 16, respectively, and the jet with the two tabs eventually entrains more mass than that with four tabs. Hence, the jet mixing for *C*4 is greater than that for *C*5. Thus, *C*4 and *C*5 are analogous to the jet manipulated by two and four tabs, respectively.

V. CONCLUSIONS

A Mach 1.4 primary jet surrounded by a sonic secondary jet is experimentally investigated to understand the mixing promoting capability of the jet under manipulation. Four different tab configurations were studied, viz., tabs placed in the primary jet (*C*2), tabs placed in the secondary jet (*C*3), and tabs placed in both primary and secondary jets with the relative orientation between them of 0° (*C*4) and 90° (*C*5). The primary jet *C*0 and coaxial jet without tab (*C*1) are also studied to understand the mixing effectiveness of the manipulated jet with that of the jet without tab. Extensive Pitot measurements are performed along the jet centerline, and the jet stream has been visualized using the shadowgraph technique in the orthogonal planes of the main jet. The following conclusions are drawn from the investigation:

1. The L_c^* of *C*1 for $U_s/U_p \geq 0.91$ is substantially elongated in comparison with that of *C*0. The elongated jet stream for *C*1 is due to two reasons. First, the secondary flow reduces the growth rate of the primary shear layer. In fact, the secondary core is not discernible for $U_s/U_p \geq 0.91$. Second, for $U_s/U_p \geq 0.91$, the strong presence of Mack wave radiation prevails between the secondary flow and the ambient fluid ($M_{cs\infty} > 1.0$, Table I). Evidently, the strong coupled behavior of the secondary flow and the primary flow results in the

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reduced growth of the shear and, hence, the elongation of the jet core. Thus, the core length is elongated with the addition of a secondary flow. Once manipulated, the L_c^* for the manipulated jets decreases. The jet mixing benefits quantified using the percentage reduction in the core length $\Delta L_{c,C1}^*$ reveal that the manipulated jet is least effective for $U_s/U_p \geq 0.91$ than $U_s/U_p \leq 0.85$. As one would expect, the presence of Mach radiation $U_s/U_p \geq 0.91$ leads to the reduced jet spread, and hence, the $\Delta L_{c,C1}^*$ reduces. This indicates that manipulating the coaxial jet in the presence of Mach wave radiation is not effective in augmenting the jet mixing. In contrast, the passive control is highly effective in reducing the core of the jet when the Mach wave radiation is eliminated.

2. The tabs employed appropriately in the primary and/or secondary jets of a coaxial jet behave as a non-circular jet emulating the characteristics of an elliptic jet. Two categories of the flow have been identified based on the tab configuration. In category 1, for *C*2, *C*3, and *C*4, the tabs bifurcate the jet in the *xy*-plane and shrink in the *xz*-plane. Hence, the jet stretches and contracts in the *xy*- and *xz*-planes, respectively. On the contrary, in category 2 for *C*5, the jet initially stretches in the *xy*-plane due to the primary tab, and at the same time, the secondary tab contracts the jet in the *xy*-plane. As a result of this, the jet spread in the *xz*-plane is reduced than that for *C*4, and hence, the jet mixing decreases for *C*5. Notably, the tab blockage for *C*4 and *C*5 is 10%, which is twice that in *C*2 and *C*3. In view of the additional 5% blockage introduced by *C*4, the increase in $\Delta L_{c,C1}^*$ achieved by *C*4 is not appreciable, which is only 20%. Thus, it is commendable to emphasize that the secondary tab may also lead to thrust loss, in addition to the thrust loss introduced by primary tabs. Hence, considering the thrust loss, it can be easily identified that the performance of *C*2 is superior to that of *C*4.
3. Tabs have been shown to reduce the noise of a supersonic jet.[49–51] Detailed acoustic measurements may be performed in the future to assess the noise benefits associated with the tabs in a co-flow jet. This would help determine how effective is the tab in reducing the noise of a co-axial jet. Finally, it would be informative to perform Particle Image Velocimetry (PIV) measurements to understand the flow physics associated with the jet mixing or noise reduction.

ACKNOWLEDGMENTS

The authors thank the anonymous referees for their constructive comments and concrete suggestions that led to significant improvements of this work. A.K.P. acknowledges the support from the Science and Engineering Research Board, India, through Grant No. EEQ/2018/000179 and would like to thank Dr. Aravindh Kumar S. M. for the fruitful discussion related to Mach wave radiation.

DATA AVAILABILITY

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to ethical restrictions.

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